



How does rewetting propagate through restored peatlands? An integrated surface–subsurface modelling analysis of water–table dynamics

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Abstract

Peatland restoration, through drainage suppression, is widely implemented to recover hydrological and ecological functions; however, the groundwater system responses governing rewetting efficiency remain insufficiently quantified across spatiotemporal scales. Here, a boreal fen is used as a process-representative system to resolve how restoration interventions, including ditch infilling and damming, reorganise catchment-scale water-table dynamics, drawing on calibrated, fully integrated three-dimensional physics-based modelling (HydroGeoSphere). Simulated restoration elevated domain-scale median water table by ~23 cm, with comparable gains (~22 cm) in nominally undisturbed areas, revealing that drainage legacy extends well beyond visibly ditched zones. Variogram analysis confirmed a shift from fragmented, drainage-controlled conditions to hydraulically connected systems with increased spatial correlation lengths. Water-table recovery propagated according to distinct hydrogeologic response regimes depending on restoration structure and subsurface properties. Low-permeability peat infillings generated strong localised responses with rapid exponential attenuation of water-table recovery (~70% decline within ~40 m), whereas dams produced broader, plateau-like responses extending up to ~100 m. Geomorphic setting further modulated recovery, with peat thickness and proximity to peat–mineral margins acting as primary controls through a threshold-like response regime. Seasonal hydroclimatic forcing modulated recovery, with wet-season response nearly doubling water-table rise relative to dry subarctic winters. These results demonstrate that drainage suppression reorganises peatland groundwater systems by enhancing lateral connectivity and controlling how hydraulic responses propagate across space. While the absolute magnitudes are site-specific, the identified controls and response patterns provide a mechanistic, transferable basis for guiding restoration in comparable boreal fens.

Keywords Wetlands · Fen hydrology · Peatland rewetting · Groundwater/surface-water relations · HydroGeoSphere

Introduction

Peatlands are among the most carbon-dense ecosystems on Earth (Joosten 2009). Although they cover only ~3% of the global land surface, peatlands store more carbon than all

terrestrial biomass combined (Beaulne et al. 2021; Strack et al. 2022), far exceeding mineral–soil ecosystems (Page et al. 2022; Winton et al. 2025). Beyond carbon sequestration, peatlands regulate catchment hydrology through water storage and delayed release, moderating floods and droughts (Shuttleworth et al. 2019; Ahmad et al. 2020), improve water quality through nutrient retention and sediment filtration (Holden et al. 2006; Haapalehto et al. 2014; Kieckbusch and Schrautzer 2007; Kadlec and Wallace 2008), and support specialised, often endangered biodiversity (D'Astous et al. 2013). However, these services have been increasingly compromised by anthropogenic disturbance. An estimated 14–20% of global peatlands have been affected by land use and management (Strack 2008), primarily through drainage, with only boreal and temperate regions accounting for ~60% of the total drained areas (Tubiello et al. 2016).

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Drainage lowers the water table (WT), accelerates peat decomposition, and increases greenhouse gas emissions, with each 1 cm decline in WT associated with an average increase of $\sim 0.3 \text{ t CO}_2\text{e ha}^{-1} \text{ yr}^{-1}$ (CO_2 equivalents, 100-year global warming potentials), mainly through enhanced CO_2 release (Evans et al. 2021; Koch et al. 2023). As a result, this extensive drainage has potentially transformed peatlands from a net carbon sink to a carbon source (Frolking et al. 2011), threatening centuries of accumulated natural carbon capital and thereby reducing ecosystem resilience to climate change (Dise 2009). Because local hydrological conditions strongly govern the severity of these impacts (Limpens et al. 2008; Strack et al. 2022), peatland restoration (primarily through rewetting) has become a central focus of climate mitigation and ecological recovery policies, including the United Nations Decade on Ecosystem Restoration (2021–2030), the European Union (EU) Nature Restoration Law with legally binding targets through 2050, and emerging national and international restoration programmes (Girkin et al. 2023; Leifeld and Menichetti 2018).

Restoration projects in boreal peatlands, often involving rewetting interventions through dam construction or ditch infill, aim to restore natural hydrological gradients aiming to reinstate the ecohydrological functions typical of undisturbed peatlands (Page and Hooijer 2016; Stachowicz et al. 2025; Karimi et al. 2025). Assessment of these project outcomes primarily relies on using before-after-control-impact (BACI) designs to monitor WT rise as a proxy for successful rewetting (Armstrong et al. 2022; Bring et al. 2022). These studies show that rewetting produces modest but ecologically meaningful rises, typically 5–9 cm, though responses can vary widely from negligible change to over 30 cm depending on site conditions (Wilson et al. 2010; Menberu et al. 2016; Holden et al. 2017; Karimi et al. 2024). The response is strongly localised, greatest within a few tens of meters of blocked ditches and decaying approximately exponentially outward, with meta-analyses reporting a $\sim 50\%$ decline within $\sim 9 \text{ m}$ and drainage impacts propagating farther still (Bring et al. 2022). The resulting improvements are spatially nonuniform, leaving cross-catchment effects uncertain (Isoaho et al. 2024).

Such projects—and the research assessing them—are usually short-term, spanning 4 to 5 years, driven by funding cycles or collaborations between academic institutions and other stakeholders (e.g. the EU LIFE projects; Finland's national rewetting programmes) (Glenk and Martin-Ortega 2018; Metsähallitus 2014). Given that pre-drainage hydrological gradients are rarely restored within the first few years (Karimi et al. 2024) and peatland ecosystems develop over decades to millennia (Young et al. 2017), the outcomes of these short-term interventions and monitoring cannot capture the full trajectory of long-term recovery (Silva et al. 2024). Moreover, most assessments rely on sparse

point-scale measurements that fail to capture peatland spatial variability, leaving the broader hydrological effectiveness of rewetting (particularly their propagation from intervention sites) insufficiently documented at spatial and temporal scales relevant to ecosystem recovery. Because these non-linear patterns reflect processes that field measurements alone cannot resolve, a mechanistic understanding is needed, which distributed hydrological modelling combined with geospatial post-processing can provide, extending beyond the spatiotemporal limitations of BACI-based studies.

Spatial hydrological analyses using models such as TOPMODEL (Goudarzi et al. 2021), SAGA (Ikkala et al. 2022), DigiBog_Hydro (Putra et al. 2022), FLUSH (Haahti et al. 2016), and MODFLOW (Rossi et al. 2014; Brandyk et al. 2016), have been applied to the northern peatland restoration (Silva et al. 2024), providing a foundation for planning interventions. However, despite decades of research, such modelling studies fall short of capturing the full complexity of restoration-driven hydrological dynamics. Many are often groundwater-only models, fail to capture fully critical surface–subsurface interactions and feedbacks, lack realistic representations of geomorphic drainage and restoration structures, provide only local-scale assessments, and are often temporally stationary. As a result, intraseasonal rewetting efficiency, lateral propagation of effects, and broader catchment-scale hydrological responses remain poorly represented in existing modelling studies, offering only a fragmented, mechanistically shallow view of restoration outcomes across such heterogeneous landscapes.

To address these knowledge gaps, a fully integrated, physically based surface–subsurface hydrological model of a boreal fen-type peatland was developed and calibrated using HydroGeoSphere (Brunner and Simmons 2012), and applied in two configurations explicitly representing the drained and restored states. Both configurations were run transiently over a full hydrologic year, generating daily spatial outputs that supported a suite of geospatial and statistical analyses designed to evaluate the following research questions: (1) How do rewetting interventions reshape the spatial and temporal dynamics of WT depth (WTD) in previously drained peatlands? (2) To what extent does restoration re-establish hydraulic connectivity, spatial coherence, and extend the effective reach of groundwater recovery? (3) How does internal peatland geomorphology regulate the lateral spread of restoration-induced WT recovery?

Material and methods

Site description

Matorovasuo is a 224-ha (2.24 km²) groundwater-fed northern aapa mire located within the 556-ha Matorova headwater

catchment in Kittilä, Northern Finland, inside the Pallas Research Area, ~170 km north of the Arctic Circle (Fig. 1). The mire lies at the southern edge of northern aapa mires, in the north-boreal vegetation zone, and forms a minerotrophic fen system receiving both surface and groundwater inputs, supported by numerous springs along its margins. The peatland drains naturally southwards through the Mato-oja stream (Outlet Q-ms; Fig. 2), while the artificial ditch network converges towards three main collector ditches (outlets Q-e1, Q-e2, and Q-w; Figs. 1b and 2); together they convey water into Lake Kivijärvi, located about 7 m below the northern part of the mire. As typical for aapa mires, the site is gently sloping and exhibits a patterned surface composed of strings (elongated hummocks) oriented perpendicular to the downslope direction, with flarks (wet depressions) occurring between them. The peatland forms a heterogeneous mosaic of wet open sedge fens, thin-peated pine-sedge fens, treed riparian zones, and scattered mineral soil islands, with vegetation including brown-moss and Sphagnum flarks, forest-moss hummocks, and dwarf shrubs such as *Betula nana*. The site has been designated as a UN Educational,

Scientific and Cultural Organization (UNESCO) ecohydrology demonstration site (2026), underscoring its relevance as a reference system for peatland hydrological restoration.

Forestry drainage began at the pine fen margins in the late 1960s–early 1970s and subsequently expanded across the peatland through an approximately 49 km ditch network (with ditch lines as close as 10 m apart), altering ~60% of the area, facilitating forest growth (Scots pine mainly); this extensively drained sector is classified here as the *disturbed* area (Fig. 1b, c). Even though the entire peatland basins were surrounded by ditches that affect the overall hydrology, the central open sedge fens parts have remained largely intact (wet and treeless) and are considered *undisturbed* area (Hultman et al. 2024; Fig. 1b).

Matorovasuo lies within the Lapland rapakivi granite region, where the bedrock is characteristically weathered and porous up to depths of several tens of meters (Hultman et al. 2024). Although the overall quality of the bedrock beneath the Matorovasuo peatlands is not well constrained, ground-penetrating radar (GPR) measurements indicate the presence of minor fractures (Hultman et al. 2024). In the

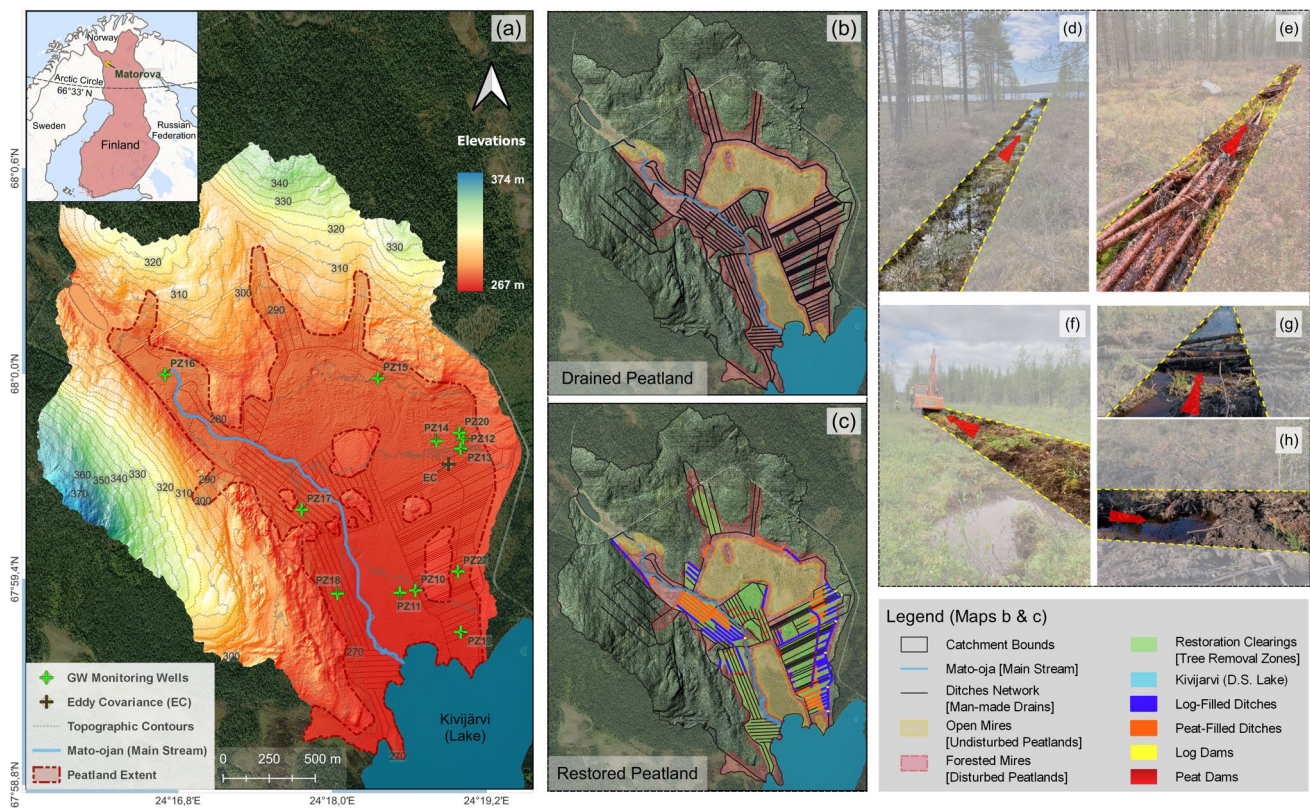


Fig. 1 Overview of the Matorovasuo mire and implemented restoration measures. (a) shows the catchment layout over a digital terrain model (DTM), with the mire boundary delineated in brown and the monitoring network indicated. (b) presents the land-cover map highlighting the pre-restoration drainage network (black lines) and internal management divisions (shaded polygons). (c) illustrates spatial

layout of the restoration plan. (d–f) provide representative field examples of drainage ditches under different conditions: (d) open (unrestored), (e) log-filled, and (f) peat-filled. (g–h) Cross-ditch structures (dams) constructed from (g) logs and (h) peat. Photograph accreditation: Omar Nimr and Anna Autio

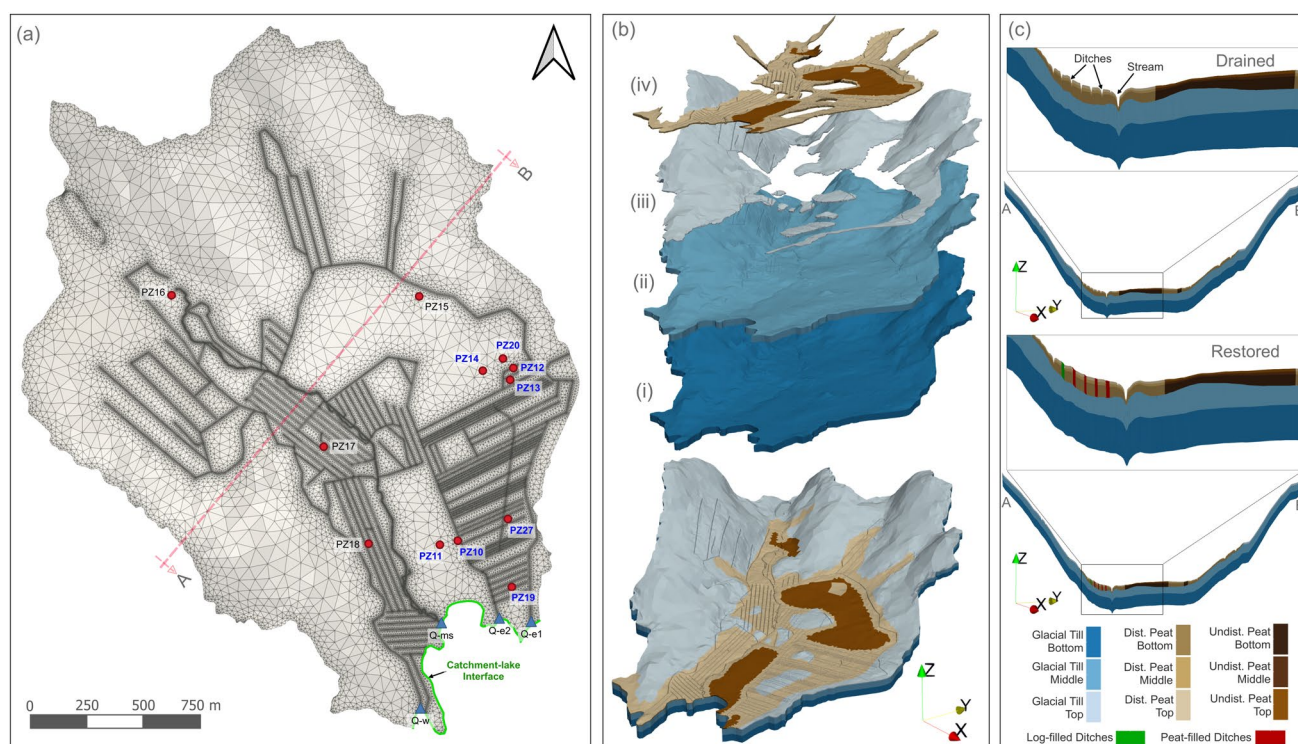


Fig. 2 Three-dimensional HydroGeoSphere model structure for the Matorova catchment. **(a)** Two-dimensional computational mesh of the model domain showing catchment stream outlets and the monitoring network; wells installed pre-restoration are labelled in blue, and wells installed post-restoration are labelled in black. **(b)** Model stratigraphy (exploded for clarity), comprising three glacial-till units, bottom (i), middle (ii), and superficial/top (iii), and the peat body (iv) resolved as three sublayers. The superficial till and peat share the uppermost

band, meeting laterally at the peat margin, giving five computational layers. **(c)** Cross-section A–B through the model domain, comparing the drained and restored setups. Ditches appear as open terrain depressions in the drained setup and as filled footprints in the restored setup, where the infill material extends through the full peat-layer thickness, which varies along the section, above the unchanged mineral layers. Colours denote the assigned material definitions (legend). Abbreviations: Disturbed (Dist.) and Undisturbed (Undist.).

surrounding uplands, bedrock is mapped as granodiorite, granite, or tholeiitic basalts, while the thickness of the weathered rapakivi layer is expected to decrease at higher elevations due to glacial erosion and to increase within the valley (Johansson and Kujansuu 2005). The surficial geology was shaped during the last glaciation and subsequent deglaciation approximately 10,000 years ago, leaving generally thin Quaternary deposits dominated by gravelly sandy till and sandy till (Johansson and Kujansuu 2005). The climate is typical for northern Finland, with a mean annual temperature of 0.4 °C (2003–2019), and mean July and January temperatures of +13.9 °C and –11.3 °C, respectively. Mean annual precipitation is 647 mm (2008–2019), 42% of which falls as snow, with snow cover lasting from mid-October to late May, reaching maximum depths of up to 105 cm in April, and winter soil freezing to ~50 cm (Marttila et al. 2021).

Restoration programme

Restoration work at Matorovasuo was carried out in 2024 under the EU LIFE PeatCarbon programme, which, to our

knowledge, represents one of the northernmost peatland restoration efforts to date. The primary objective was to re-establish the peatland's hydrological regime, particularly water balance, internal flow pathways, and WT elevations, to conditions approximating those prior to forestry drainage. Historical aerial imagery and detailed field surveys were used to identify original flow routes, which guided the design of restoration measures (Hultman et al. 2024). Restoration activities began in winter 2024 with tree removal conducted under frozen conditions to minimise soil disturbance, followed by ditch blocking and manual earthworks during summer 2024. All operations were supervised by the landowner, Metsähallitus (Finnish Forest and Park Service).

Post-drainage tree encroachment, mainly *Betula nana* and Scots pine, was selectively removed on the basis of tree age, condition, and comparison with a 1957 aerial photograph, with the aim of returning evapotranspiration and, therefore, catchment-scale water balance to near-natural levels. Harvested trees and associated logging residues were repurposed as restoration material. Approximately 6.8

km of ditches were filled using logs placed longitudinally to form log-filled ditches (Fig. 1e), providing roughness, reducing flow velocity, trapping sediments, and supporting *Sphagnum* recolonisation. An additional 4.1 km of ditches were infilled using collapsed and compacted peat from ditch margins, forming peat-filled ditches (Fig. 1f). In total, 92 transverse dams, seven log dams and 85 peat dams, were constructed (Fig. 1g, h), with both types concentrated along the two major north–south ditches.

A particularly influential drainage feature is a large north–south collector ditch located about 300 m east of Mato-oja in the southern part of the peatland, flowing out at Q-e2 (Fig. 2a). Before restoration, this ditch intercepted and diverted water from the eastern and northern parts directly to Lake Kivijärvi, partially preventing lateral flow to feed the adjacent ridge fen on its west side. Blocking this collector ditch, together with another collector flowing out at Q-e1 and redirecting their inflows back towards the ridge fen was considered a key measure for restoring hydrological functions in the southern peatland (Fig. 1c).

Data acquisition

WTD was monitored using 12 IoT-enabled groundwater wells (PZ10–PZ20 and PZ27) distributed across the mire. Each well was equipped with pressure sensors (Decentlab DL-PR26 and Caproc AQUA-26) measuring WT levels and temperature, with data transmitted via LoRaWAN. Eight wells were installed in late 2022 (pre-restoration), and four additional wells (PZ15–PZ18) during and after restoration in 2024 (Figs. 1a and 2a).

Meteorological forcing data were compiled from two automatic weather stations operated by the Finnish Meteorological Institute (FMI): Kittilä Matorova and Kittilä Kenttäröva. Both stations are located within a 2 km radius of the central low-lying area of the Matorovasuo peatland. Hourly observations spanning three full years (2021–2023) were aggregated to daily values, including maximum and minimum air temperature, relative humidity, wind speed, solar radiation, snow depth, and precipitation. These variables were used to calculate reference evapotranspiration (ET_0) using the UN Food and Agriculture Organization (FAO) Penman–Monteith method (Allen et al. 1998) and served as atmospheric forcing for the hydrological model. Precipitation was partitioned into rainfall and snowfall, with snowfall defined as precipitation occurring when snow depth exceeded zero.

Actual evapotranspiration (AET) was independently estimated using latent heat flux (LE) measurements from an eddy covariance (EC) tower located within the peatland (Baldocchi 2003; Wu et al. 2010) (Fig. 1a). LE data was collected from June 2023 to January 2024 at 30-min intervals

and aggregated to daily values. Negative LE values, commonly associated with condensation, dew formation, or freezing conditions, were set to zero prior to aggregation to represent net evapotranspiration only. Daily AET (mm) was calculated as:

$$AET_{\Delta t} = \frac{LE}{\lambda} \Delta t \quad (1)$$

where LE is the latent heat flux ($W\ m^{-2}$), Δt the aggregation interval (s), and λ the temperature-dependent latent heat of vaporisation ($J\ kg^{-1}$), calculated following Allen et al. (1998):

$$\lambda(T) = 2.501 \times 10^6 - 2.361 \times 10^3 T \quad (2)$$

with T representing mean air temperature ($^{\circ}C$).

Topographic data were derived from a 2 m-resolution DTM (airborne laser scanning, 2018; N2000 vertical reference), publicly available from the National Land Survey of Finland (National Land Survey of Finland 2018) (Fig. 1a). Peat depth was manually measured at 72 locations in June 2024 using a steel probe, recorded via RTK-GPS, and interpolated to a continuous surface of peat-mineral soil interface, assuming zero thickness at peatland boundaries (Fig. S2 of the electronic supplementary material (ESM)).

Depth to bedrock at the Matorova catchment was characterised using GPR. Surveys were conducted using 50 and 100 MHz antennas to balance penetration depth and resolution. A winter 2024 survey employed a snowmobile-towed system that simultaneously carried both antennas, enabling the acquisition of shallow, high-resolution data and deeper profiles in a single pass. Full survey details are reported in Jadoon (2025). Interpreted GPR-derived bedrock depths were interpolated within the catchment domain to produce a continuous bedrock depth surface, with depths ranging from ~3 m at upland divides to ~14 m in low-lying riparian areas (Fig. 2b).

Modelling approach

Integrated surface–subsurface flow model

HydroGeoSphere (Aquanty, Inc, 2015; Brunner and Simmons 2012), a fully integrated, physically based surface–subsurface flow model was employed to simulate the hydrological dynamics of the Matorova catchment (5.6 km²; Fig. 2). The planar domain was discretized into 114,147 triangular elements and 57,571 nodes, with finer resolution (down to 0.5 m) near streams, ditches, and restoration interventions, and coarser elements (up to 60 m) in forested and open mire areas far from the features. The triangulation was well-conditioned, with a minimum interior angle of 30° and a mean reciprocal aspect ratio of 0.75 ± 0.10 (unity denoting

an equilateral element) and no elements below 0.45. Surface and subsurface properties were discretized into parsimonious (piecewise) parameter zones mapped onto the 2D computational mesh, while the vertical domain was divided into five stratified layers, to capture variability in soil properties within the profile.

Surface and subsurface flow were solved simultaneously within a single, globally implicit control-volume finite-element system (Therrien et al. 2010). Three-dimensional variably saturated subsurface flow was described by the Richards equation, and two-dimensional overland and in-channel flow by the diffusion-wave approximation of the depth-averaged Saint–Venant equations, with flow resistance and micro-topographic storage represented through Manning roughness and rill-storage parameters (Table S1 of the ESM). The two domains were coupled through a dual-node approach, in which the exchange flux at each surface node was computed as a first-order function of the head difference across a thin interface layer; a coupling length of 1×10^{-3} m was applied to all overland units, giving tight, near-continuous coupling between the surface and the underlying peat.

Two model configurations were developed: a drained (pre-restoration) setup and a restored (post-restoration) setup. To preserve strict comparability, both used an identical computational mesh (the same horizontal node positions, element faces, and vertical layering) and differed only in the digital terrain model (DTM) and in the hydraulic properties assigned at restoration footprints. In the drained DTM, ditches were incorporated as explicit terrain depressions incised to a constant 1 m below the local, spatially variable ground surface, with a width of 1.5 m, hereafter referred to as the *ditched DTM*. In the restored DTM, the mapped ditch footprints were filled back to adjacent bank level following the restoration plan (Fig. 1c). The infill material properties were assigned across the full, spatially variable thickness of the peat layer within each footprint, while the underlying glacial till layers retained their drained-setup parameterisation. Dams were imposed analogously by raising ditch-bottom nodes to adjacent bank level over the crest footprint (crest widths ~3–5 m). Ditch and dam footprints were resolved by the locally refined mesh (element edges down to 0.5 m, roughly three elements across the 1.5 m ditch width) and were represented as DTM-elevation and material-property zones on this shared mesh. Hydraulic behaviour was parameterised according to construction materials; peat infills were informed by studies on peat bunds used in restoration (Menberu et al. 2021; Putra et al. 2022), while log infills were parameterised using values reported for large-wood hydraulic structures (Manners et al. 2007; Addy and Wilkinson 2019).

Subsurface geology was informed by 2024 GPR surveys, peat-depth measurements, boreholes, field observations,

and regional datasets (GTK 2018). The vertical domain was discretized into five computational layers. The peat body was resolved into three sublayers, with relative thickness proportions of 0.5, 0.5, and 2 from top to bottom, capturing the well-documented rapid vertical changes in peat hydraulic properties (Ingram 1978; Holden 2005). Beneath the peat, two glacial-till layers (bottom and middle) continued to the bedrock elevation at the base of the 3D grid. A third, surficial (top) till occupied the uppermost band across the basin where peat was absent, meeting the peat laterally at its margin. Disturbed and undisturbed areas were assigned distinct hydraulic properties horizontally and vertically (Fig. 2). The overland flow (i.e., surface) domain was discretized into four land-cover types, including open “undisturbed” and forested “disturbed” peatland, streams/ditches, and natural forest areas (Fig. 1), each assigned specific overland flow parameters (Table S1 of the ESM).

Boundary conditions were assigned to the surface and subsurface domains independently. The catchment–Lake Kivijärvi interface (Fig. 2) carried a first-type (Dirichlet) specified-head condition on the subsurface domain, fixing the saturated hydraulic head at lake stage (267.75 m, N2000) and thereby controlling groundwater discharge to the lake; co-aligned along the same shoreline, the surface domain was assigned a critical-depth condition permitting free overland and channel outflow. Precipitation, partitioned into rainfall and snowfall (see the “Data acquisition” section), was imposed as a daily, time-varying Neumann flux boundary on the model top; snowfall accumulated in a surface snowpack and was released as meltwater through a calibrated temperature-index routine, so that the timing of water delivery to the surface domain reflected snow accumulation and melt rather than instantaneous precipitation. The remaining lateral boundaries and the model base were no-flow. Streamflow in the Mato-oja channel and the ditch network was therefore nowhere prescribed; it emerged entirely from the coupled solution, and surface water–groundwater exchange along streams and ditches was resolved dynamically, with reaches gaining or losing water locally according to the evolving head gradient between the channel and the peat. Actual evapotranspiration was computed, by the model, as the realised fraction of potential evapotranspiration constrained by energy availability and soil moisture, combining surface evaporation with subsurface evaporation and transpiration following Kristensen and Jensen (1975) and Allen et al. (1998). Forested areas undergoing tree removal were assigned modified ET properties in the restored setup to account for logging impact (Fig. 1).

Soil freeze–thaw and the formation of stream and ditch ice were not represented as explicit thermal phase-change processes. Frost-limited infiltration and ephemeral ice damming of channels were therefore not resolved; consistent with the annual water-balance and long-term-equilibrium

focus of the study, their principal effect would be to shift the timing rather than the magnitude of the simulated rewetting response.

Calibration procedure

The parsimonious, physics-based structure of the model enabled the assignment of physically meaningful initial parameters within zone-based parameterisations. Initial values and bounds for peat hydraulic properties (hydraulic conductivity, anisotropy, porosity, and van Genuchten parameters; Tables S1 and S2 of the ESM) were primarily guided by Menberu et al. (2021) for pristine (undisturbed) peat and forestry-drained peat, while till parameters were informed by the nearby Pallaslompola catchment with a comparable geological setting and formation history (Autio et al. 2023).

Calibration was conducted using a sequential, multi-stage approach, applied to the drained model configuration. First, a manual (trial-and-error) steady-state calibration was conducted to reproduce mean WT elevations at 12 monitoring wells (Fig. 2), adjusting hydraulic conductivities and retention parameters to achieve realistic groundwater storage and regional WT patterns. Although four of these wells were installed during or shortly after restoration implementation (black-labelled in Fig. 2), their average WT elevations were assumed to be minimally affected after a couple of months from restoration, therefore allowing their information content to be used as approximate constraints for steady-state WTDs.

Second, transient calibration was carried out for 2022–2023 using the PESTPP-GLM algorithm (Doherty 2015). A total of 31 parameters were estimated (Table S1 of the ESM) using nine observation groups, comprising WTD from eight pre-restoration wells and time-varying actual evapotranspiration derived from eddy covariance measurements. A single multivariate objective function was minimised, with observation weights applied to balance contributions across groups and prevent domination by variables with larger magnitudes or differing units (Table S3 of the ESM). Regularisation was employed within the PEST++ framework to stabilise the inverse problem and limit ill-posedness (e.g. nonuniqueness). This included running PESTPP-GLM in regularisation mode using Tikhonov regularisation (Tikhonov and Arsenin 1977), and the singular value decomposition (SVD) utility available in PEST++ as described in Aster et al. (2018), together with the inherently parsimonious piecewise-uniform parameter zoning.

To assess how strongly the observation network constrains the estimated parameters, and thus the confidence in the reported metrics, the calibration was complemented with a linear (first-order, second-moment; FOSM) parameter-uncertainty and identifiability analysis of the calibration Jacobian, performed within the pyEMU framework (White

et al. 2016). For each process group, the prior-to-posterior reduction in parameter variance and the parameter identifiability obtained from singular value decomposition of the weighted Jacobian was quantified (Doherty and Hunt 2009), together with the effective dimensionality of the inverse problem. The full methodology, equations, and group-level results are provided in Supplementary text S1; Figs. S3 and S4; and Table S4 of the ESM.

Model performance was evaluated using kernel density plots of observed versus simulated WTD across multiple wells to assess agreement across the full dynamic range. For actual evapotranspiration and snow depth, conventional scatter plots were sufficient, as each variable was derived from a single observational dataset. A 1:1 reference line and ordinary least squares regression were included to assess bias and dispersion. Performance was quantified using multiple complementary metrics, including Kling–Gupta efficiency (KGE), root mean square error (RMSE), Nash–Sutcliffe efficiency (NSE), and the coefficient of determination (R^2).

Following calibration, production simulations were conducted separately for the drained and restored model configurations over a full hydrologic year (1 October 2022 to 30 September 2023); the restored-configuration outputs represent the simulated long-term hydrological equilibrium under restored boundary conditions, not a near-term transient response. Climatic forcing was applied at a daily time step, and spatially distributed outputs for all relevant HydroGeoSphere flow variables, primarily including groundwater elevation, WTD, and surface saturation, were generated at daily intervals. Initial conditions for these simulations were obtained from approximately two-year spin-up runs (January 2021 to October 2022), each initialised from steady-state conditions. Steady state was established using quasi-transient simulations, in which time-invariant forcings were applied in transient mode until changes in simulated water storages (ΔS) became negligible (i.e., converged to near zero).

Variogram analysis of water-table depth

Spatial dependence and autocorrelation of mean annual WTD under drained and restored conditions were quantified using experimental variogram analysis. The variogram (semivariogram) describes how the dissimilarity between pairs of values increases with their separation distance, and is therefore well suited to characterising the spatial structure of the simulated WT. For a lag distance h , the experimental semivariance was estimated using the Matheron estimator,

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2 \quad (3)$$

where $Z(x_i)$ is the mean annual WTD at location x_i and $N(h)$ is the number of point pairs separated by approximately h . Each experimental variogram was modelled with an exponential function,

$$\gamma(h) = c_0 + c \left[1 - \exp\left(-\frac{h}{r}\right) \right] \quad (4)$$

where c_0 is the nugget, c the partial sill (the spatially structured variance above the nugget), and r the range parameter (the e-folding distance, at which the model attains $1 - 1/e \approx 63\%$ of the structured sill); the model approaches its sill, $S = c_0 + c$, asymptotically. Parameters were estimated by nonlinear least squares. The effective correlation range, a , reported throughout, was defined as the lag at which the model reaches 95% of the structured sill, $a = -r \ln(0.05) \approx 3r$. The exponential form was selected because it reproduces the monotonic, asymptotic decay of spatial correlation expected for a diffusion-controlled field such as the WT. The three parameters carry direct hydrological meaning: the nugget (c_0) reflects unresolved sub-grid variability and short-range noise in WTD; the sill ($S = c_0 + c$) is the total spatial variance of WTD, i.e. the overall magnitude of WT heterogeneity; and the effective range (a) is the characteristic distance over which WTD values remain spatially correlated. A longer range indicates that the WT varies coherently over greater distances; this is interpreted as a correlation length as a proxy for, rather than a direct measure of, hydraulic connectivity.

Gridded WTD outputs were downsampled to a uniform 10 m grid, from which a fixed random subsample of 20,000 cells (random seed 42) was drawn per condition for the all-pairs variogram computation, keeping the pairwise calculation tractable while preserving spatial structure. The drained and restored grids were collocated. Omnidirectional experimental variograms were computed separately for each condition using all point pairs in planar Cartesian coordinates, grouped into 50 lag classes with the maximum lag set to the 95th percentile of all inter-point distances; the exponential model was then fitted by nonlinear least squares to obtain c_0 , c , and r , from which the sill $S = c_0 + c$ and the effective range a were derived.

Directional variograms were subsequently computed to assess potential anisotropy in spatial dependence. Semivariances were calculated for point pairs with azimuths within $\pm 22.5^\circ$ of the NW–SE (135°) and NE–SW (45°) directions, chosen to represent orientations approximately parallel and perpendicular to the dominant slope-driven drainage of the catchment. For each direction and management condition, semivariances were binned into 50 lag

classes up to the 95th percentile of directional distances, and exponential models were fitted independently using the same criteria as the omnidirectional analysis.

To further evaluate anisotropy, two-dimensional variogram maps were generated for drained and restored WTD. Pairwise semivariances were aggregated into a Δx – Δy lag space using symmetric bins extending to the 95th percentile of inter-point distances. These maps visualised semivariance as a function of lag magnitude and orientation, enabling qualitative identification of anisotropic patterns, and were displayed using a common semivariance colour scale with contour overlays. All variogram analyses were implemented in Python; the experimental omnidirectional and directional variograms and the variogram maps were computed with custom NumPy routines using the Matheron estimator, and the exponential model was fitted by nonlinear least squares (`scipy.optimize.curve_fit`) under non-negativity constraints.

Geospatial correlation and regression of restoration-induced water-table rise

The spatial propagation of restoration effects on WT rise was quantified using the difference between restored and drained annual mean WT (Δ WT) at 0.5 m resolution. Analyses were restricted to peat-covered cells, defined as locations with non-zero peat depth and excluding mineral islands within the peatland extent. Restoration features, including peat-filled ditches, log-filled ditches, and dams, served as reference polylines, from which Euclidean distances to each peat cell were calculated, generating ~ 8 million pointwise measurements. For each feature type, Δ WT was then correlated with distance, and values were binned into 40 intervals to reduce noise from the dense pointwise dataset and to emphasise the underlying trend. Within each bin, the mean Δ WT was computed, and whiskers representing ± 1 standard deviation were used to illustrate local variability. Binned trends were fitted with regression models appropriate to the feature type (e.g. exponential decay for ditches), with equations and goodness-of-fit metrics reported. A parallel analysis assessed natural geomorphic controls on Δ WT, including peat thickness, slope of peat formation (calculated as the spatial gradient of peat thickness), and distance to the peat–mineral interface. Metrics were raster-aligned, correlated with Δ WT, binned, and fitted with nonlinear regression models (polynomial or exponential) as appropriate. Outliers beyond the 1–99th percentile were excluded to avoid spurious correlations. This workflow enabled quantification of both the lateral propagation of restoration effects and the geomorphological controls shaping Δ WT responses across the peatland.

Results

Model calibration performance

Calibration, guided by the conceptual understanding of the Matorova system, assigned lower permeability to glacial till at hillslopes and beneath the peatland relative to peat (Table S2 and Fig. S1 of the ESM). Peat permeability decreased with depth, and disturbed layers were calibrated to be more permeable than undisturbed, ensuring realistic groundwater storage and WT dynamics. Calibrated van Genuchten parameters yield steeper soil water retention curves for till, reflecting narrower available water ranges than for peat. Optimised parameters, including saturated hydraulic conductivity, van Genuchten water retention parameters, porosity, and overland flow and evapotranspiration coefficients, are summarised in Tables S1 and S2 of the ESM.

WTD from the transient simulations yielded a large number of simulation–observation pairs, and model performance was evaluated using kernel density estimation plots across all wells (Fig. 3). The joint density of WTD is concentrated near the 1:1 line, indicating good agreement between observed and simulated WT. High-density contours occur at both low/negative WTDs (near or above ground) and higher WTDs (0.5–0.75 m), indicating that most data points cluster within these ranges. Where, at the same time, these ranges also exhibit the widest spread in kernel density estimation contours, reflecting greater dispersion in the observed–simulated relationship across wells. Aggregated metrics of the time-variable WTDs indicate overall good model performance: KGE 0.61, RMSE 0.19 m, NSE 0.56, and R^2 0.61.

Well-specific time series (Fig. 3) further indicate that the model captures a broad range of local groundwater behaviours, with most wells exhibiting R^2 values between 0.5 and 0.7. It reproduces shallow and/or negative WTDs indicative of ponding or upwelling conditions (spring zones), such as those observed at PZ12, and deeper WT behaviour at nearby wells (e.g. PZ13–PZ14), demonstrating its ability to resolve spatially heterogeneous groundwater processes. Some wells exhibit weaker performance indices; however, their simulated time series still reproduce the dominant temporal trends, with deviations largely attributable to systematic biases, particularly during dry-season periods (e.g. PZ10, PZ11, and PZ19).

The model successfully reproduces system-scale daily actual evapotranspiration, with KGE 0.62, RMSE 0.72, NSE 0.60, and R^2 0.63 (Fig. 4), though slight positive bias occurs at low AET rates. Snowpack dynamics are accurately reproduced following calibration of the degree-day melt factor ($2.6 \text{ mm } ^\circ\text{C}^{-1} \text{ day}^{-1}$) and threshold

temperature ($0.1 ^\circ\text{C}$), with performance metrics KGE 0.78, RMSE 11.53 cm, NSE 0.90, and R^2 0.96 (Fig. 5), though minor overestimation occurs at peak snow depths likely due to the simplified assumption of constant snow density. Together, the combined calibration of groundwater storage (i.e. WTD), atmospheric fluxes (i.e. AET), and snowpack dynamics ensures a physically consistent, closed representation of the catchment water balance.

Beyond goodness-of-fit, the parameter-uncertainty and identifiability analysis confirmed that the calibration is well constrained where it matters most. Calibration reduced the variance of the peat saturated hydraulic conductivities (the parameters exerting first-order control on the simulated WT response) by 95–98%, and these conductivities accounted for the largest share of the resolved parameter information; the till and overland-flow parameters were moderately constrained (≈ 77 –84%), and the evapotranspiration parameters least. Across all process groups, the observation network resolved approximately six independent parameter directions—substantially fewer than the nominal number of calibrated parameters—indicating that the inverse problem is well posed and that the reported metrics are governed by a small set of strongly identified parameters (Figs. S3 and S4; Table S4 of the ESM).

Spatiotemporal dynamics of water-table response under drained and restored conditions

WTD dynamics, spatially constrained to the peatland extent, were analysed over a complete water year (October 2022–September 2023) for the two management scenarios (drained and restored) and are presented using monthly violin plots (Fig. 6). Each violin represents the combined distribution of daily WTD values across all peatland grid cells within a given month, and is split to allow direct side-by-side comparison between the two scenarios.

Across both scenarios, wider sections of the violins occur at near-surface and negative WTD values, corresponding to surface water presence in open ditches and spring-fed areas, and around ~ 1 m depth, particularly in riparian zones and inter-ditch areas strongly influenced by drainage. These widened regions indicate depth ranges with a higher spatial–temporal frequency of occurrence.

In contrast to the drained configuration, the restored scenario exhibits a consistent downwards shift of the violin distributions and their associated quartile lines (Q1, median, and Q3) across all months, indicating a systematic rise of the WT induced by rewetting interventions. Beyond this overall elevation change, restoration also alters the shape of the WTD distributions. Whereas the drained scenario shows pronounced skewness towards deeper WTD values, manifested as persistent “bulging” around ~ 1 m, the restored

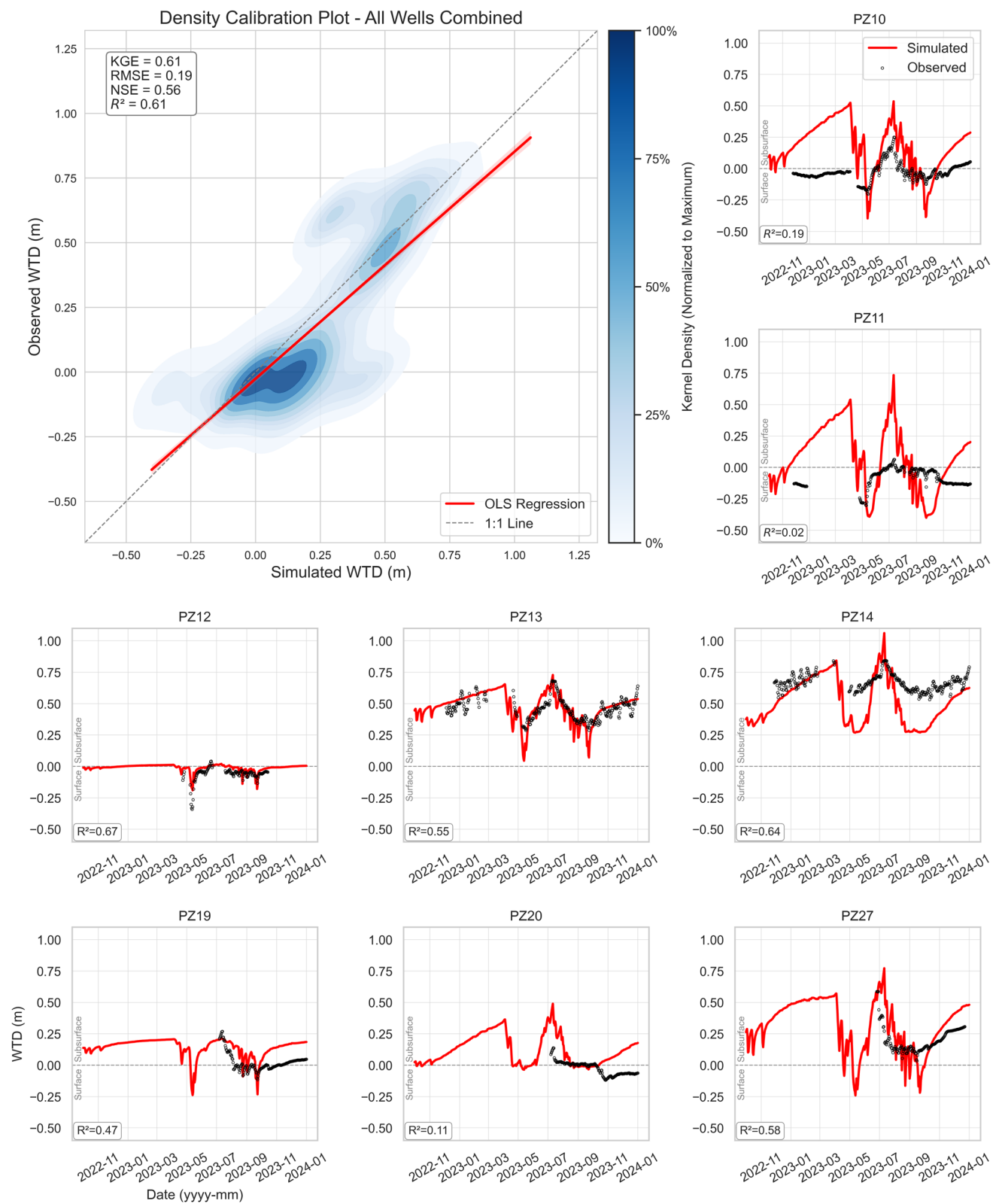


Fig. 3 Transient-state calibration of groundwater dynamics. Simulated versus measured WTD at eight groundwater observation wells within the Matorvasuo domain for the drained setup; wells correspond to those installed pre-restoration and labelled in blue in Fig. 2

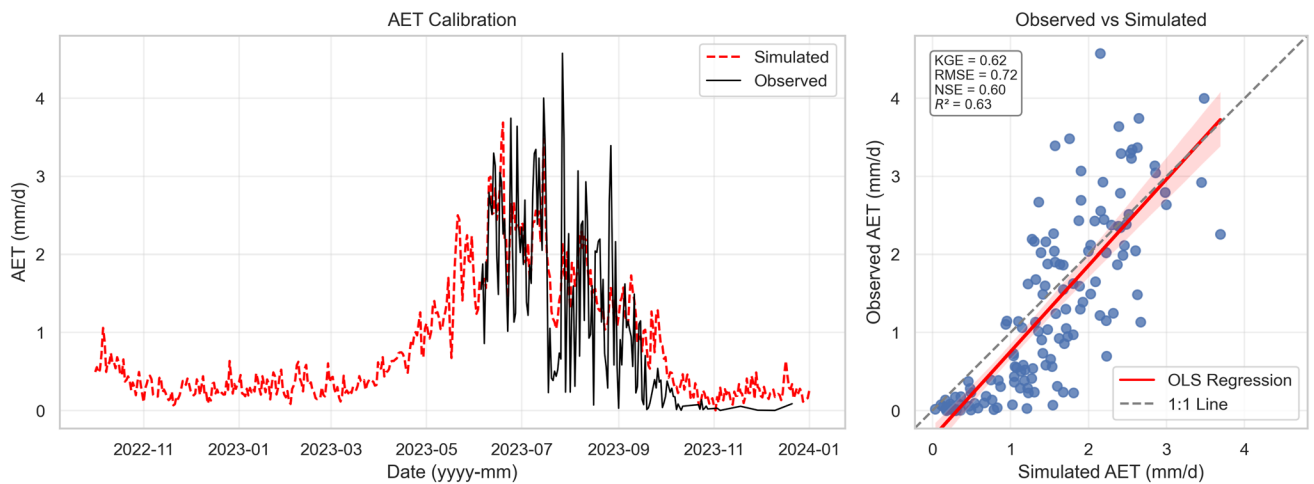


Fig. 4 Transient-state calibration of actual evapotranspiration. Simulated versus observed actual evapotranspiration rates within the Matorvasuo domain for the drained setup

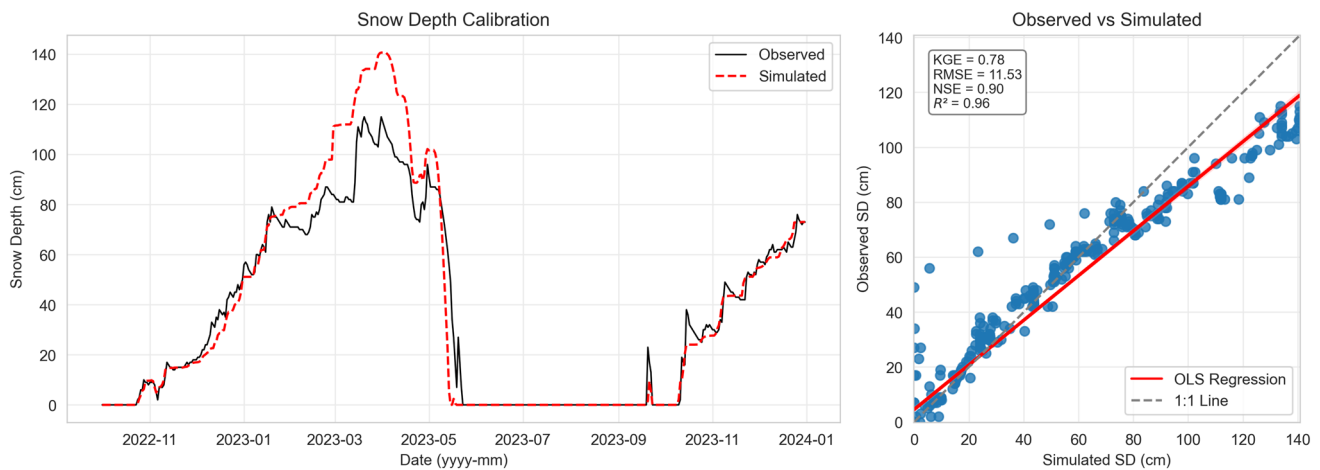


Fig. 5 Transient-state calibration of snow dynamics. Simulated versus observed snow depth within the Matorvasuo domain for the drained setup

scenario displays a more vertically uniform distribution across shallower depth ranges (Fig. 6). Additionally, it is observed that the interquartile ranges of monthly WTDs are consistently narrower under restored conditions (year-average IQR 79 cm) than under drained conditions (94 cm). Collectively, these changes reflect a redistribution of simulated WT conditions in both space and time, consistent with a re-established, domain-scale groundwater coverage rather than localised zones of deep drawdown.

The overlaid median lines further elucidate temporal WTD dynamics, revealing pronounced inter-seasonal variability (Fig. 6). Both management scenarios exhibit clear and consistent seasonal patterns, with shallower WTDs during wetter periods, most notably following spring snow-melt in May (23 cm for restored vs. 47 cm for drained) and

during periods of intense autumn rainfall in September (21 vs. 45 cm), and deeper WTDs during drier phases, particularly over winter, reaching maximum depths in March (67 vs. 84 cm), and again in mid-summer (July 59 vs. 82 cm) (Fig. 6).

To quantitatively assess the temporal efficiency of the rewetting programme in elevating WT levels at the peatland domain-scale, a daily difference in simulated WT elevations between the restored and drained configurations was computed using domain-wide median WTD values. This metric, expressed as ΔWT , provides a direct and intuitive index of restoration efficiency through time (Fig. 7), smoothed with a 15-day moving average to highlight seasonal trends. The resulting ΔWT time series remains positive throughout the hydrological year, confirming that restoration enhances the

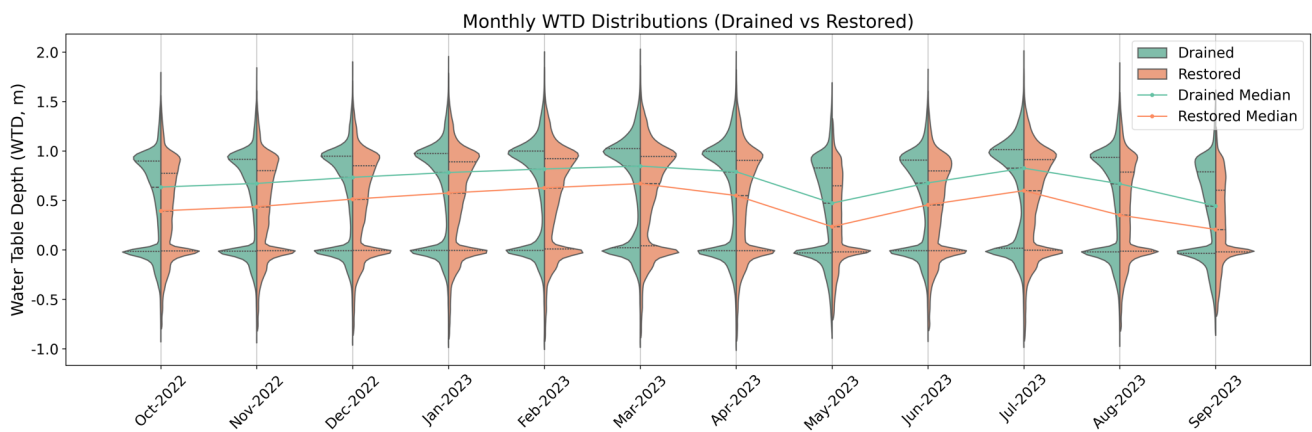


Fig. 6 Monthly distributions of daily WTD across the Matorvasuo peatland for drained and restored scenarios over a full hydrologic year (October 2022–September 2023). Each violin is split, with the drained scenario on the left and the restored scenario on the right, and displays the full range of WTD values across all peatland grid cells,

based on ~1.7 million data points (30 days $\times$ ~57,000 surface nodes). Dashed lines mark the first quartile, median, and third quartile for each month, and the connected median lines show seasonal WTD dynamics across the year

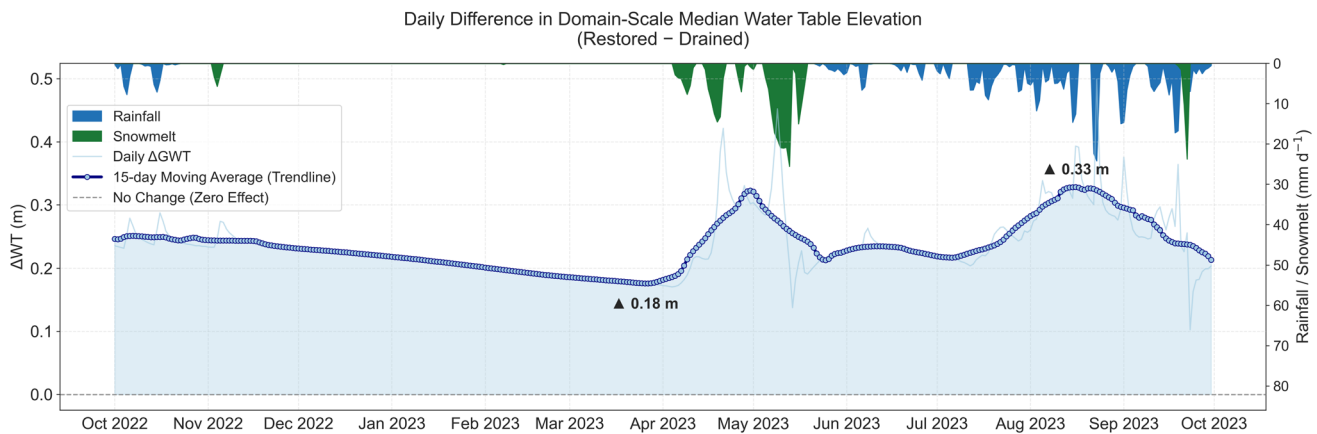


Fig. 7 Time series of the daily difference in median WT elevations (Δ WT) between restored and drained scenarios across the Matorvasuo peatland (October 2022–September 2023). Δ WT was calculated as the difference in domain-scale median WT elevations between the two scenarios. A 15-day moving average is applied to

smooth short-term variability and emphasise inter-seasonal trends. Positive values indicate higher WT elevations under the restored configuration. Catchment-scale influxes are shown above the time series: rainfall is based on observed time-varying data, and snowmelt is extracted from modelled snow balance outputs

simulated WT levels under all prevailing conditions. However, its magnitude varies seasonally, peaking at ~33 cm during spring snowmelt and autumn rainfall, declining to ~18 cm in late winter, and maintaining ~25 cm over summer. This seasonal modulation highlights the strong interaction between restoration measures and hydroclimatic forcing, with the greatest WT response during wet periods.

The spatial response of groundwater to restoration was quantified as the difference in annual mean simulated WTD between restored and drained configurations for each grid cell (Fig. 8a). Negative values (blue shades) indicate a WT rise induced by restoration, while positive values (red shades) denote lowering of the WT; transparent areas show

negligible change. The most pronounced increases in WT elevation occur in areas directly affected by ditch-infilling restoration interventions, particularly where peat-filled ditches were implemented, as well as upstream of dam structures. However, the spatial pattern overall is dominated by negative values (i.e. blue shades), indicating a widespread rise in the WT and demonstrating effectiveness of the restoration measures extending beyond their immediate footprints.

Substantial WT rises are also observed in undisturbed peatlands, including the southern-central and northeastern open mires, where no direct restoration measures were implemented. In the northeastern open mire, two localised

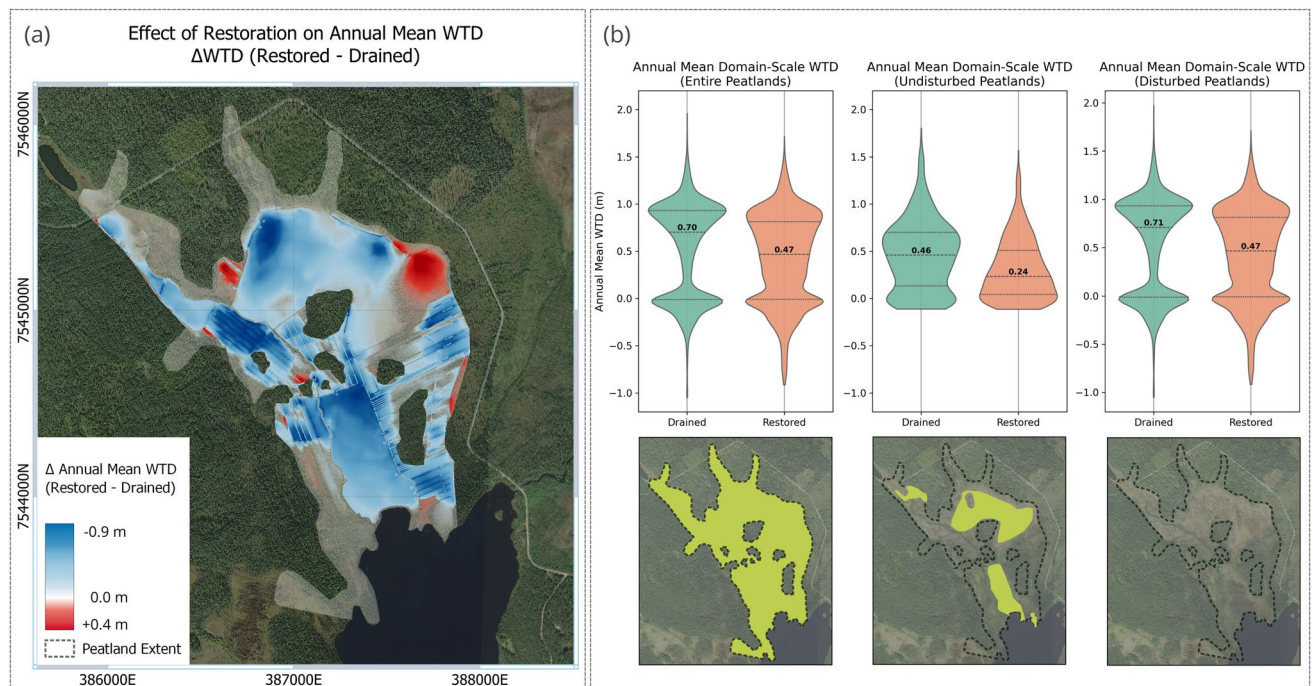


Fig. 8 Spatial and domain-scale response of groundwater to restoration across the Matorovasuo peatland. **(a)** Map of the difference in annual mean WTD between restored and drained scenarios for each model grid cell. Blue shades indicate a rise in the WT (negative WTD difference), red shades indicate local decreases, and transparent

areas denote negligible change. **(b)** Violin plots summarising the distribution of annual mean WTD across the entire peatland, disturbed peatland, and undisturbed peatland, showing the full range of grid-cell values for both scenarios

zones of particularly strong WT rise (strong blue shades) are evident near marginal ditch infilling locations, suggesting indirect hydrological influence of drainage blockage in reducing lateral groundwater losses and enhancing local storage. Similarly, in the southern-central open mire, a pronounced increase in WT elevation is observed in its north-eastern corner, likely associated with intensive damming along the Q-e2 collector drain, which historically drained the eastern portion of this mire.

Localised decreases in WT elevation (red shades) are observed at a limited number of sites, indicating that restoration can also induce spatially heterogeneous and occasionally adverse groundwater responses. The largest contiguous area of WT decline is located in the northeastern open mire adjacent to an unfilled segment of the marginal ditch. A common characteristic of these negatively affected areas is their proximity to the peat–mineral soil interface, where redirected groundwater fluxes and altered storage dynamics may locally enhance drainage. In contrast, the southwestern part of the peatland (west of the Mato-oja stream) and the two northern finger-shaped extensions exhibit negligible changes in WTD, consistent with the absence of restoration interventions in these areas.

Simulated annual mean WTD for both management scenarios is summarised using violin plots that represent

the full distribution of time–space grid-cell values, thereby capturing the integrated, domain-scale effect of restoration over the hydrological year (Fig. 8b). Violin plots are shown separately for the entire peatland, disturbed peatland, and undisturbed peatland to isolate the differential response of these landscape units to restoration interventions. For the entire peatland, the median annual mean WTD is 70 cm under drained conditions and 47 cm under restored conditions, indicating a substantial overall rise in the WT following restoration. This shift is reflected in the shape of the violin plots, where the restored scenario shows a more vertically uniform density, indicating a broader and more evenly distributed range of WTDs, whereas the drained scenario exhibits a pronounced widening at greater depths, reflecting a clustering of observations towards deeper WTDs. The disturbed peatland shows a similar magnitude and redistribution, whereas the undisturbed peatland's median WTD decreases from 46 to 24 cm, with the WT shifting towards near-surface conditions, highlighting pronounced restoration effects even in areas without direct interventions.

Spatial structure of water-table depth

Omnidirectional variograms revealed clear spatial autocorrelation in mean simulated annual WTD for both drained

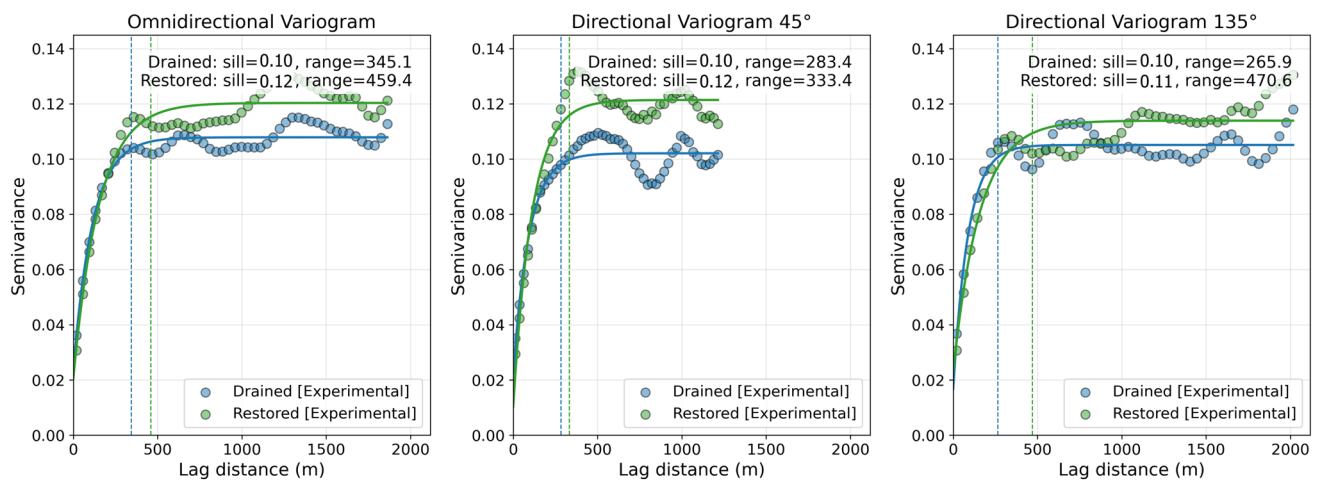


Fig. 9 Spatial autocorrelation of mean annual WTD across the Matorvasuo peatland. Omnidirectional variograms show overall spatial correlation, while directional variograms at 135 and 45° represent the major and cross slope-driven drainage directions, respectively

and restored conditions (Fig. 9). The drained WTD exhibited a sill of 0.10 and an effective correlation range of 345 m, whereas restored conditions showed a slightly higher sill of 0.12 and a longer range of 459 m, indicating stronger spatial variability and more extensive spatial correlation following restoration. Nugget values were low and comparable between scenarios (~ 0.02), suggesting minimal microscale variability or noise.

Directional variograms highlighted anisotropic patterns linked to the catchment's slope-driven drainage and ditch network alignment (Fig. 9). Along the 45° orientation (perpendicular to the overall catchment slope-driven drainage but parallel to dominant southern ditches), drained WTD had a sill of 0.10 and a range of 283 m, while restored WTD showed a higher sill of 0.12 and a range of 333 m. Along 135° (parallel to slope-driven drainage), drained WTD exhibited a sill of 0.10 and a range of 265 m, whereas restored WTD maintained a comparable sill of (0.11) but with a substantially longer correlation range of 471 m. These patterns indicate that restoration preferentially increases the spatial correlation of WTD, particularly along the principal drainage direction, while maintaining comparable variability magnitudes (sill) across both directions.

Variogram maps further elucidated directional differences in the spatial structure of mean annual WTD between drained and restored conditions (Fig. 10). Drained conditions exhibited lower semivariance at short-to-intermediate lags, particularly along the y-direction, indicating stronger local-scale spatial continuity of WTD. This pattern is consistent with hydrologic homogenisation imposed by artificial drainage, in which closely spaced ditches constrain WT gradients and promote locally coherent drawdown. At larger lag distances, however, the drained variogram map showed

higher semivariance values relative to the restored condition, especially along the x-direction. This increase in long-range semivariance suggests enhanced spatial decoupling of WTD across the catchment under drained conditions, likely reflecting the cumulative influence of ditch-aligned flow pathways and longitudinal gradients imposed by the drainage network. In contrast, the restored condition displayed more moderate semivariance at large lags, consistent with a reduction in long-range hydraulic contrasts and a partial re-establishment of catchment-scale hydrologic connectivity.

Lateral spread of restoration-induced water-table recovery

Correlation analysis between Δ WT (i.e., restoration-induced WT rise) and distance from restoration measures revealed distinct spatial propagation patterns for each feature type (Fig. 11). Peat-filled ditches exhibited a pronounced exponential decay in Δ WT, with bin-mean values decreasing from ~ 0.50 m at the ditch edge to a near-constant value of 0.15 m at 40 m distance ($\approx 70\%$ attenuation). Standard deviation across bins (i.e. whiskers) remained approximately constant at 0.4 m, reflecting local variability in WT response to restoration both along and across individual ditches at equivalent distances from the feature. The exponential decay model fitted the binned mean remarkably well ($R^2 0.98$), confirming the rapid lateral attenuation of the restoration effect.

Similar exponential behaviour was observed for dams, although the decay was less steep. Δ WT started at a maximum bin-mean of 0.35 m at the dam location and reached a plateau of 0.16 m at ~ 100 m, corresponding to a 55% reduction (from the maximum rise at the structure). Responses

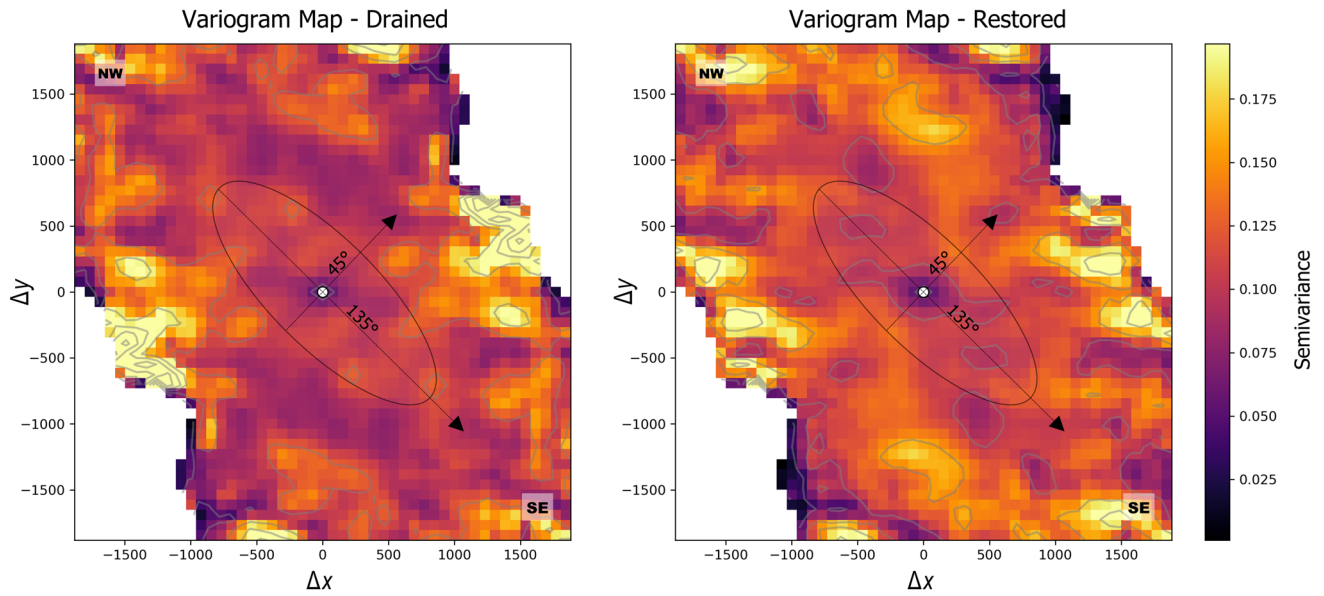


Fig. 10 Variogram maps of mean annual WTD for drained and restored scenarios, showing spatial structure differences. Arrows indicate 135° and 45° directions corresponding to major and cross slope-driven drainage axes

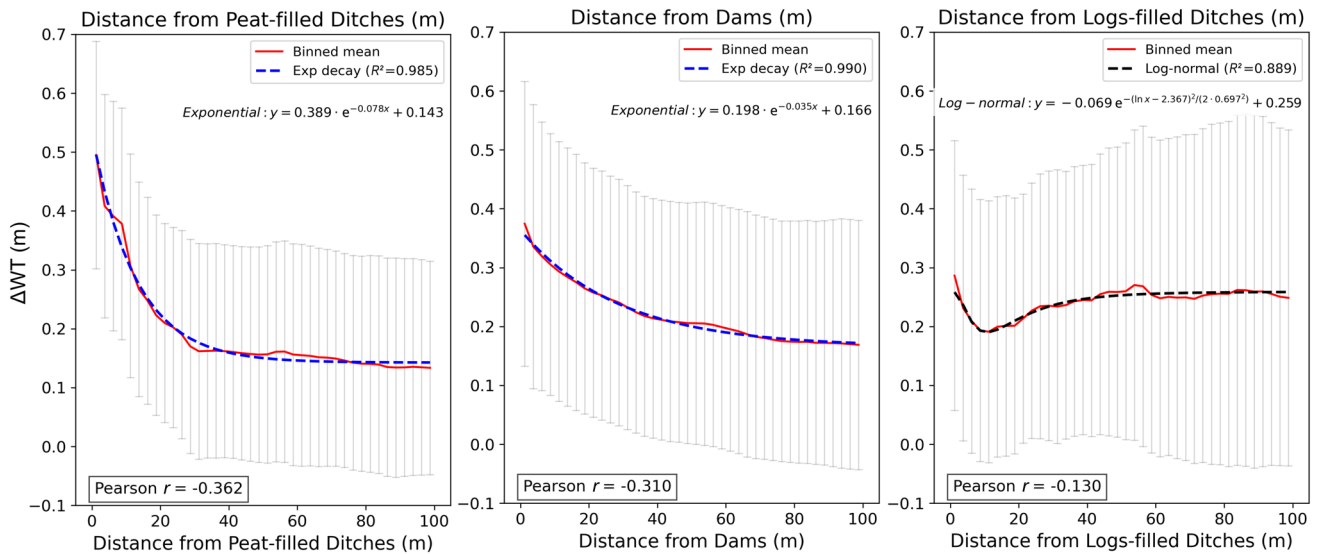


Fig. 11 Lateral propagation of restoration-induced WT rise (Δ WT) from different restoration features across the Matorvasuo peatland. Binned mean values and variability (whiskers) are shown, with fitted nonlinear models illustrating the distance-dependent effect of each

feature. The fitted-model equation is annotated on each panel, where y is Δ WT (m) and x is distance from the feature (m). Pearson correlation coefficients (r) assess fit across all individual data points; R^2 values indicate the model fit to the binned means

around dams exhibited a larger and more consistent standard deviation of ~ 0.5 m, indicating higher variability but longer-range influence compared to peat-filled ditches.

Log-filled ditches displayed a distinct, non-monotonic response best described by a log-normal-shaped regression

($R^2 0.89$). Δ WT initially declined from ~ 0.30 to ~ 0.20 m within 10 m of the ditch, followed by an increase towards a stable plateau of ~ 0.25 m beyond ~ 30 m, indicating a superimposed response in which short-range local attenuation and broader-scale redistribution mechanisms overlap spatially.

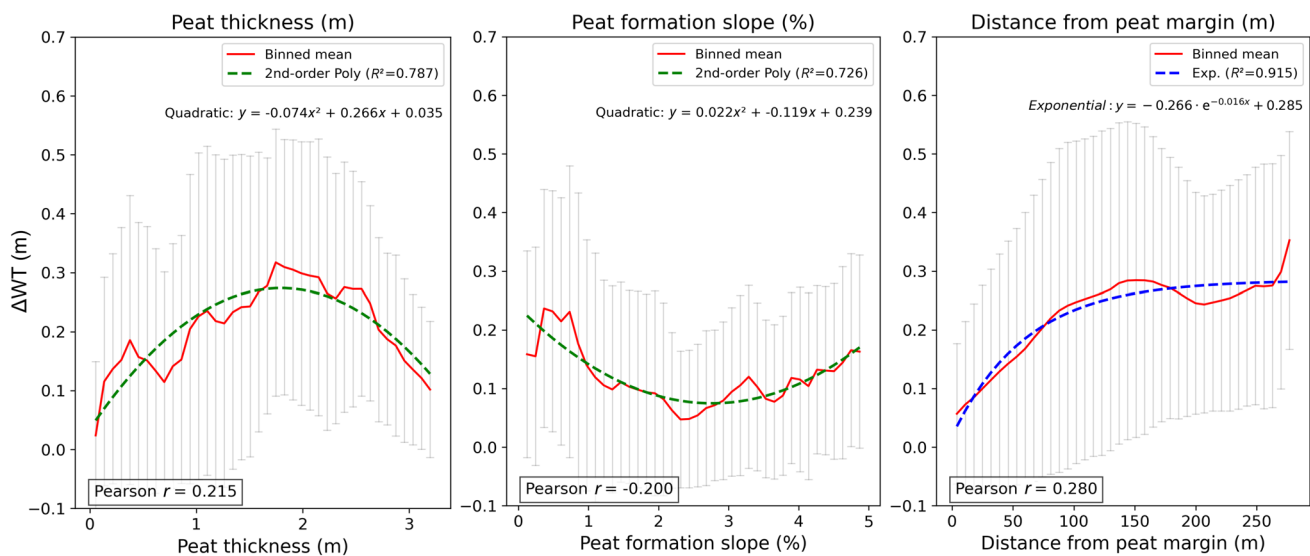


Fig. 12 Influence of peatland geomorphology on restoration-induced WT rise (Δ WT) across the Matorvasuo peatland. Binned means and variability (whiskers) are shown for Δ WT as a function of peat thickness, peat formation slope, and distance from the peat–mineral interface, with fitted nonlinear models illustrating the relationships. The

fitted-model equation is annotated on each panel, where y is Δ WT (m) and x is the variable on that panel's horizontal axis. Pearson correlation coefficients (r) assess model fit across all individual data points, whereas R^2 values assess fit to the binned-mean data

Geomorphological controls on restoration-induced water-table recovery

Correlation analysis between Δ WT and selected peatland geomorphological metrics revealed strong, nonlinear controls on restoration outcomes (Fig. 12). Peat thickness exhibits a pronounced quadratic relationship with Δ WT (R^2 0.79), with groundwater rise increasing up to a tipping point at ~ 1.8 m peat thickness, where binned-mean Δ WT peaks at ~ 0.30 m, followed by a decline at greater thicknesses. Standard deviation increased towards the tipping point, peaking at ~ 0.5 m, and decreased away from it to ~ 0.2 m, indicating greater variability in restoration response at intermediate peat thicknesses.

Peat formation slope (gradient of peat thickness) shows an inverse quadratic relationship with Δ WT (R^2 0.73), with minimum groundwater response near the slope tipping point of $\sim 3\%$ and higher Δ WT values at both low and high slope magnitudes. Variability in Δ WT was generally lower than for peat thickness, with narrower standard deviations near the tipping point and increased spread at both flat and steep slopes.

Distance from the peat–mineral interface exerted a strong control on Δ WT, following an exponential growth relationship (R^2 0.92). Δ WT was near zero at the peat margin and increased with distance towards the peatland interior, reaching a plateau of approximately 0.29 m beyond ~ 150 m from the margin. Although the relationship showed a strong fit, variability was substantial, with standard deviation peaking to ~ 0.6 m at intermediate distances (100–150 m), indicating heterogeneous restoration responses within the peatland interior.

Discussion

Synthesis and comparison with previous studies

Peatland rewetting, through ditch blocking and drainage suppression, typically shifts catchment-scale hydrology from an efficient drainage network back into a retention-dominated system (Karimi et al. 2024). This transition commonly induces a rapid but spatially heterogeneous rise in WT, with larger increases occurring where pre-restoration drawdown was strong and baseline WTDs were deep, and more modest changes where peatlands were lightly drained or already wet (Holden et al. 2011; Haapalehto et al. 2014; Menberu et al. 2016; Bring et al. 2022; Isoaho et al. 2024; Stachowicz et al. 2025).

The modelling results support this expectation, showing a substantial rise in WTs following rewetting through combined ditch infilling and damming interventions. At the site scale, simulated rewetting increased the annual mean domain-scale WT by ~ 23 cm relative to drained conditions (Fig. 8b), assuming long-term hydrological equilibrium. Notably, a comparable increase (~ 22 cm on average) was observed within peatland areas classified as undisturbed, with local rises exceeding 40 cm (Fig. 8a). In the context of drainage asymmetry (van der Kamp and Hayashi 2009; Holden et al. 2011), these responses indicate that historical drainage effects extended well beyond visibly ditched areas and that restoration can induce substantial recovery in regions previously assumed to be unaffected.

This pattern is particularly relevant for sloping fen systems, such as Matorovasuo, where ditches oriented perpendicular to slope and concentrated along peatland margins can strongly disrupt lateral inflow, isolate downslope peat from up-gradient surface water and groundwater sources, and propagate drawdown far into the peatland interior (Chimner et al. 2017; Isokangas et al. 2017). The results demonstrate that ditch blocking effectively re-establishes lateral connectivity of shallow groundwater and sheet flow across the hillslope, enabling recovery in far-field areas. This behaviour is consistent with earlier observations by Schimelpfenig et al. (2014) and underscores the potential of restoration interventions to influence hydrology far beyond the immediate intervention zones.

The findings, generally, are in line with a broad body of BACI-based field studies and hydrological modelling that report post-restoration WT rises ranging from several centimeters to a few decimeters (Shantz and Price 2006; Wilson et al. 2010; Haapalehto et al. 2011, 2014; Menberu et al. 2016; Bring et al. 2022; Isoaho et al. 2024; Karimi et al. 2024; Stachowicz et al. 2025). Site-average groundwater responses of this magnitude have been reported in ditch-blocked minerotrophic fens in southern Finland, where rewetting raised mean WT levels from deeply drained conditions to near-surface states, with average increases of ~35 cm persisting for up to a decade (Haapalehto et al. 2011), and similarly in block-cut bogs in Canada following peat-dam rewetting (Ketcheson and Price 2011). Other studies report smaller but still ecologically meaningful increases of ~5–10 cm when averaged across broader monitoring networks or multiple sites (Menberu et al. 2016; Karimi et al. 2024; Stachowicz et al. 2025). While on the other hand, some studies report little or no detectable response in the first years following rewetting (Holden et al. 2017; Shuttleworth et al. 2019). This apparent discrepancy is largely a matter of spatial scale rather than inconsistency. Fine-scale hydrodynamic models (~20 ha) capture localised head buildup behind dams and bunds, reporting fairly large water-level increases, up to ~70 cm in tropical peatlands (Putra et al. 2022). By contrast, a landscape-scale simulation covering tens of thousands of hectares reported a domain-mean groundwater rise of only ~1–2 cm, as much of the domain lies hydraulically distant from drainage features, and the raised head propagation is limited by peat conductivity contrasts and microtopography (Urzainki et al. 2023). Remote-sensing-based modelling further supports this spatial concentration of effects, showing heterogeneous groundwater rises (~4.7–8.8 cm) that decay rapidly with distance from restoration measures (Isoaho et al. 2024).

This wide inter-site variability highlights the strongly site-specific nature of restoration outcomes, heavily contextualised by baseline conditions such as differences in climatic forcing, peat type and thickness, the intensity

and legacy of past drainage, hydrological state, restoration extent, and landscape connectivity (Haapalehto et al. 2011, 2014; Menberu et al. 2016; Holden et al. 2017; Swindles et al. 2025). As a result, single-value WTD magnitudes offer only a partial, and potentially misleading, representation of system response in the peatland complex. Here, a diagnostic model-based framework was adopted to elucidate how restoration reorganises groundwater dynamics in space and time, including the spatial redistribution of WT states, seasonal modulation under varying hydroclimatic forcing, and lateral propagation from intervention features. This framing emphasises that restoration outcomes are defined not merely by groundwater rise, but by the mechanisms through which restoration alters hydraulic gradients, connectivity, and storage across the peatland system, yielding insights that are more transferable than absolute values alone. Beyond corroborating this consensus, the study addresses key gaps by applying fully integrated, three-dimensional physics-based modelling (a first in the context of peatland restoration) to reasonably quantify the lateral and temporal propagation of restoration-induced groundwater rise and to mechanistically resolve the geomorphological and connectivity controls governing spatial recovery.

From fragmentation to connectivity: Domain-scale reorganisation of water table under restoration

Monthly WTD distributions (Fig. 6) show that rewetting not only elevates WT levels but reorganises hydrodynamics across the peatland. The consistent upward shift in WT under restored conditions demonstrates a sustained rise throughout the year, while reduced skewness toward deep WTs indicates suppression of localised drawdown zones typical for drained conditions. Narrower interquartile range further reflects dampened groundwater variability. Together, these changes point to a re-established lateral hydraulic connectivity, rather than isolated head buildup near individual interventions.

Variogram analysis provides independent, scale-explicit evidence that restoration restructures WTD spatial coherence at the fen scale (in Figs. 9 and 10). Longer omnidirectional correlation ranges under restored conditions (459 vs. 345 m) indicate that WTD fluctuations remain spatially correlated over larger distances, consistent with a shift from ditch-controlled flow towards a more continuous, topography- and peat-property-controlled groundwater system (Kennedy and Price 2004; Haahti et al. 2016). This post-rewetting reorganisation is most pronounced along the principal slope-driven drainage axis, where the marked extension of correlation length (135°; 471 vs. 265 m) further suggests partial re-establishment of longitudinal connectivity and attenuation of sharp hydraulic discontinuities imposed by the former ditch network (Jaenicke et al. 2010). In contrast, the shorter

correlation ranges (Fig. 9) and stronger anisotropy (Fig. 10), under drained conditions, reflect spatial fragmentation with locally coherent drawdown and long-range decoupling. Low and comparable nugget effects imply that restoration primarily alters mesoscale flow organisation and lateral coupling rather than microscale heterogeneities, while the modest increase in sill reflects a reorganised spatial variability rather than a fundamental change in point-scale variance. Collectively, these patterns signal a shift towards a more integrated, “pristine-like” hydrological regime characterised by enhanced spatial coherence and reduced long-range decoupling (Haapalehto et al. 2014; Holden et al. 2006). Such geospatial analysis is possible only with physics-based simulations, as in this study due to the lack of field measurement capabilities at such high spatial resolution.

The redistribution and stabilisation of WTDs following restoration arises from a cascade of interacting hydromechanical feedbacks within the peat matrix (Haapalehto et al. 2014; Nimr et al. 2024; Waddington et al. 2015). Blocking high-conductivity drainage pathways forces lateral flow into lower-permeability peat, reducing the connected thickness of hydraulically efficient near-surface layers and lowering system-scale effective transmissivity (Waddington et al. 2015). Rising WTs saturate previously unsaturated acrotelm layers and partially reconnect them with the catotelm, altering both vertical and lateral water redistribution. An elevated WT also flattens hydraulic gradients towards former drainage features, slowing lateral fluxes while promoting more uniform water storage (Waddington et al. 2015; Holden et al. 2017). The physics-based model, however, captures these principal hydrological responses to restoration; it does not represent other autogenic hydromechanical feedbacks, such as peat-matrix swelling or “mire breathing” which enhance near-surface storage and retention during high-intensity recharge (Waddington et al. 2015). Omitting these feedbacks likely biases the model towards underestimating domain-scale retention and overestimating WTD responsiveness to external forcing.

Seasonal expression of restoration efficiency in rewetted peatlands

Model results show that restoration elevates the baseline WT year-round, yet the magnitude of this effect is strongly modulated by seasonal hydroclimatic forcing (Fig. 7). Consistent with boreal and temperate studies, ditch blocking and bunds induce a magnitude of a rise that is larger and spatially more extensive during wet periods, while responses are muted in dry, high-evaporation seasons (Waddington et al. 2015; Putra et al. 2022; Karimi et al. 2024; Stachowicz et al. 2025). During snowmelt and autumn rainfall, high water availability saturates the peatland, driving shallow WT that actively engages with restoration structures. These structures

function as secondary hydraulic media, blocking preferential flow paths, generating local upstream head buildup, and promoting controlled percolation into adjacent peat matrix. This generates disproportionately high WT rises relative to drained conditions. In contrast, during dry periods, under low-supply conditions, the deep WT may remain below the intervention thresholds, and the head may never rise enough to activate flow into the surrounding peat, reducing efficiency. Moreover, high hydraulic gradient between the pool behind a dam and the dry area below it can force water to seep much faster through the dam or preferentially flow along relict ditch margins, limiting the persistence of surface-adjacent WT (Holden et al. 2017; Stachowicz et al. 2025).

Between spring and autumn peaks, restoration efficiency remains relatively high and stable (Fig. 7), coinciding with the seasonal window most vulnerable to drought under projected climate change (Chan et al. 2023). This reflects enhanced peatland storage efficiency: water captured during high-recharge events is retained in the saturated profile and gradually released, buffering the system during drier spells, in line with observed post-restoration dynamics (Karimi et al. 2024).

Seasonal contrasts in restoration efficiency are also modulated by evolving of autogenic hydrological feedback linking WT dynamics to peat decomposition state, vegetation productivity, peat deformation, and surface resistance and energy balance, among other feedbacks documented in the literature (Waddington et al. 2015; Nugent et al. 2018). These self-regulating feedbacks progressively alter peat properties—hydraulic conductivity, capillary connectivity, bulk density, acrotelm structure, and macropore formation—shaping how restored peatlands translate inputs into WT rise (Waddington et al. 2015; Howie and Hebda 2018; Nijp et al. 2019; Ahmad et al. 2020). While the present modelling framework effectively captures the hydrological responses to altered boundary conditions and engineered storage, these internally driven, seasonally evolving peat-hydromechanical feedbacks remain unresolved, consequently limiting interpretations on the restoration efficiency.

Hydraulic controls on the lateral propagation of restoration-induced water-table recovery

The observed lateral spread of restoration-induced groundwater rise (Δ WT) reflects both the design of interventions and the hydraulic properties of their construction materials (Fig. 11), fundamentally altering the geometry and connectivity of groundwater redistribution within the peat body.

Peat-filled ditches, modelled with very low permeability, induced a sharp, strongly localised rise in WT, with Δ WT declining by $\sim 70\%$ within ~ 40 m (Fig. 11). Acting as intense hydrological plugs (Bring et al. 2022), the low permeability

of the peat infill impedes lateral flow, imposing strong hydraulic resistance, causing water to accumulate immediately upstream of the ditch, and generating a steep local hydraulic gradient. This configuration effectively reverses drainage-induced drawdown at the source but dissipates rapidly into the surrounding peat matrix, explaining the steep exponential decay observed (Haapalehto et al. 2011, 2014). The excellent fit of the exponential model (R^2 0.98) indicates that groundwater re-equilibration is governed by short-range lateral head diffusion in the surrounding peat, imposed by the low-conductivity infill rather than by long-range redistribution, and is fully consistent with field and meta-analysis evidence for strongly localised effects (Armstrong et al. 2009; Bring et al. 2022).

Peat dams exhibited a less steep decay and a substantially longer radius of influence, reaching a plateau at ~100 m from the structure (Fig. 11). Unlike peat-filled ditches, dams do not uniformly restore hydraulic resistance along the drainage network; instead, they create discrete zones of impoundment that redistribute water both laterally and longitudinally (Fig. 8a). They primarily function by raising upstream water levels, generating backwater effects in open channels and lateral head gradients that propagate through connected peat and shallow flow pathways (Waddington et al. 2015; Karimi et al. 2024). The consistently larger standard deviation in Δ WT around dams (~0.5 m) suggests that their effectiveness is likely modulated by connectivity to adjacent ditches and local network structure. Where hydraulic continuity is high, dams can transmit rewetting signals over considerable distances; where it is low, responses remain patchy (Ramchunder et al. 2012).

The non-monotonic pattern observed around log-filled ditches (Fig. 11) indicates a fundamentally superposition of multiple restoration effects rather than single-structure control. The weak near-field response, followed by a rapid increase and a distal plateau, indicates that individual log-filled ditches exert limited hydraulic control on their own. Their high permeability renders them hydraulically “leaky” (Haapalehto et al. 2014), allowing water to easily pass through the fill rather than accumulate locally, limiting local WT rise. Therefore, while a single leaky log-fill may be ineffective, a network of these features provides increased surface and subsurface roughness, slowing water across the catchment and leading to a synergistic “retention state” (Holden et al. 2017).

These patterns highlight the interplay between hydraulic resistance and restoration design: low-permeability infillings deliver strong, local rewetting, high-permeability features require cumulative effects to generate measurable impact, and dams facilitate broader-scale groundwater redistribution, albeit with greater spatial variability. Therefore, these results support viewing peatland rewetting as a mosaic of overlapping hydraulic perturbations, whose net effect depends on

feature density, spatial arrangement, and material properties. Furthermore, the fact that all features eventually plateau (at 0.15 to 0.25 m) rather than returning to zero suggests that restoration establishes a baseline level recovery, inherently emerging from the superimposed restoration effect by all single restoration features (Elo et al. 2024). This convergence complicates the isolation of single-feature lateral influence and underscores the need for future design-oriented studies disentangling individual versus landscape-scale restoration effects.

Geomorphic controls on restoration-induced water-table recovery

The observed nonlinear relationships between restoration-induced WT rise (Δ WT) and peatland geomorphology (Fig. 12) indicate that restoration effectiveness is strongly conditioned by the internal architecture of the peat body and its hydraulic coupling to the surrounding mineral substrate. Peat thickness exhibits a pronounced quadratic relationship with Δ WT (R^2 0.79). The speculated reason is that thin peat provides minimal storage capacity, and restoration-induced head increases are rapidly dissipated into the underlying low-permeability mineral substrate. As peat thickness increases toward the interior, storage capacity and lateral transmissivity increase, enabling restoration measures to achieve substantial and spatially coherent groundwater recovery. Beyond the tipping point (~1.8 m), however, further increases in peat thickness, likely introduce greater vertical heterogeneity and internal hydraulic damping, dispersing restoration-induced head increases over a larger volume and reducing the observable Δ WT signal (Belyea and Clymo 2001; Holden 2005). The peak in variability near this threshold suggests that intermediate-thickness peatlands represent a transitional regime (i.e., storage-limited to diffusion-limited) in which small structural differences exert a disproportionate influence on outcomes.

While large-scale meta-analyses suggest that peat depth is not always a statistically dominant “moderator” when looking at average outcomes across all sites (Bring et al. 2022), a high-resolution field study by Menberu et al. (2016) shows that the peat thickness strongly influences the magnitude and rate of WTD recovery. Menberu et al. (2016) report an inverse relationship between peat thickness and the vertical WT response, with thinner peat layers exhibiting the largest mean WTD recovery following restoration, a pattern likely evident with systems where the underlying mineral soil is more permeable than the peat. In such systems, the WT at thinner peatlands may be more responsive to groundwater upwelling from the underlying mineral soil, whereas very thick peat layers can act as a “confining layer” that isolates the surface hydrology from the wider catchment (Rossi et al. 2014).

The exponential increase in Δ WT with distance from the peat–mineral interface indicates a progressive hydraulic decoupling from marginal boundary effects (Fig. 12). Near boundaries, thin peat overlying low-permeability mineral substrate, which acts as the system’s regional groundwater source, provides insufficient transmissivity to counter the transient outward flux. During high-flux periods, water accumulates in the peat interior faster than it can infiltrate the mineral substrate, driving lateral flow towards the boundary, dissipating WT rise and temporarily turning the margin into a local hydraulic sink. Moving towards the peatland interior, increasing distance from this boundary reduces lateral leakage, so cumulative confinement by the surrounding mineral envelope enhances water retention, allowing restoration signals to integrate spatially and reach a plateau beyond ~ 150 m (Fig. 12). While previous studies have recognised the influence of peat–mineral boundaries on WT dynamics (Rossi et al. 2014; Nimr et al. 2024), this study is the first to quantitatively model and resolve the nonlinear spatial relationship between boundary proximity and restoration-induced groundwater recovery.

The inverse-quadratic relationship between peat formation slope and restoration-induced groundwater rise (Fig. 12) suggests a geomorphic optimum around $\sim 3\%$ slope, at which peatlands are already strongly hydrologically buffered and gains from restoration are limited. This aligns with studies showing that groundwater-fed peatlands that are rewetted can maintain a buffered WT and reduced responsiveness once strong internal connectivity is established (Menberu et al. 2016; Ahmad et al. 2020; Lambert et al. 2022). Around such an “optimal” formation slope, the peat body is well connected hydrologically, leaving less potential Δ WT to recover through rewetting (Nijp et al. 2019; Ahmad et al. 2020). By contrast, very flat sectors can exhibit large Δ WT increases when stagnant or ponded areas are reconnected, whereas steeper sectors can respond strongly when efficient drainage pathways are interrupted, consistent with work on canal blocking, controlled drainage, and altered flow paths in degraded peatlands (Urzainki et al. 2020; Ikkala et al. 2022; Salla et al. 2024). Consequently, both flatter and steeper formation slopes show larger and more variable groundwater responses, in line with topography-controlled flow dispersion, peat volume-change feedbacks, and slope-dependent connectivity to drains and surrounding aquifers reported in other peatlands (Hokanson et al. 2020; Autio et al. 2020; Li et al. 2024).

Overall, the findings show that hydrological responses to rewetting are process-general yet configuration-specific; meaning that the direction and nature of change are governed by fundamental mechanisms, such as storage redistribution, transmissivity reduction, lateral connectivity, and boundary conditions, while the magnitude, spatial

reach, and seasonal expression depend on restoration density, measures placement, and geomorphic context. As such, the results present a transferable, mechanistic framework for anticipating how restoration can be optimised across diverse peatland settings, thereby informing prioritised design decisions.

Parameter identifiability and robustness of the reported metrics

Confidence in the reported metrics rests on the calibrated parameters themselves. Although the inverse problem involves a large number of parameters, the linear uncertainty and identifiability analysis (Supplementary text S1 of the ESM) shows it to be far better posed than this nominal count implies. The observation network resolves only about six independent parameter directions, and these are concentrated in the peat saturated hydraulic conductivities that exert first-order control on the simulated WT response. The remaining parameters exert progressively weaker influence in proportion to how strongly the observations resolve them, so that the least-resolved parameters carry little leverage on the groundwater metrics reported here. The headline outcomes are, moreover, relative quantities: the domain-scale rise (~ 23 cm) is a drained-to-restored contrast computed on a single, identically parameterised model, so parameter errors that bias absolute heads act on both configurations and largely cancel in the difference. Likewise, the characteristic influence distances (e.g. ~ 40 m around infilled ditches, ~ 100 m around dams) and the post-restoration correlation range (459 m) are controlled primarily by domain geometry and the contrast in conductivity between open and infilled features, structural attributes that are comparatively insensitive to the absolute calibrated values. Because the reliability of a model prediction is governed by its dependence on the parameter directions that calibration actually resolves (Moore and Doherty 2005; White et al. 2014), the reported metrics, anchored in the well-resolved peat conductivities, are therefore robust to the residual parameter uncertainty that calibration does not eliminate. A complete probabilistic characterisation, propagating the full posterior parameter distribution into predictive ensembles, was beyond the present scope and is a natural next step, particularly once post-restoration monitoring enables joint constraint of the restored configuration.

Scale and process limitations of the modelling framework

The interpretation of these results must acknowledge a few scale- and process-bound constraints inherent to the modelling framework. First, the restored model configuration was not independently validated against post-restoration

field observations, and it should be emphasised that such a validation would presently be conceptually inappropriate rather than merely data-limited. The restored configuration is solved under steady restored boundary conditions with time-invariant hydraulic parameters and therefore represents the long-term hydrological equilibrium of the rewetted system. All wells of the monitoring network have begun to register post-restoration WT signals, but these records capture only the initial transient of a system that typically requires years to decades to equilibrate after rewetting (Wilson et al. 2010; Holden et al. 2011; Haapalehto et al. 2011; Karimi et al. 2024). A quantitative comparison between an equilibrium projection and such an early transient record would not constitute a valid test, since agreement would be fortuitous and disagreement expected. Importantly, the restored configuration is not an independently fitted model; it retains the entire calibrated parameterisation of the drained baseline (see the “Model calibration performance” section) and differs only in the digital terrain model and in the material properties at ditch and dam footprints. The restored scenario outputs should therefore be interpreted as physically consistent projections of long-term hydrological equilibrium under restored boundary conditions (a constrained perturbation of a calibrated model), rather than as predictions of near-term hydrogeological response. Future monitoring campaigns at Matorovasuo, once post-restoration equilibration is achieved, should be used to revisit and formally test these projections.

Second, groundwater responses are simulated using time-invariant hydraulic parameterisations. This precludes the representation of autogenic ecohydrological feedbacks—including peat matrix swelling (“mire breathing”), progressive macropore occlusion by Sphagnum regrowth, and changes in acrotelm structure driven by vegetation recovery (Waddington et al. 2015; Howie and Hebda 2018). Because these feedbacks generally act to enhance near-surface water retention and reduce WTD responsiveness to external forcing, their omission likely causes the model to overestimate the short-term magnitude of WT rise and its sensitivity to hydroclimatic variability. Notably, however, the simulated domain-scale gains of approximately 23 cm fall within the range consistently reported by field-based studies across comparable boreal systems (Haapalehto et al. 2014; Menberu et al. 2016; Stachowicz et al. 2025), providing partial corroboration of their physical plausibility despite this potential upwards bias. The simulated restoration-induced gains in the WT should therefore be understood as representing an upper-bound scenario under the current hydraulic regime, with actual long-term recovery trajectories likely modulated by these progressive ecohydrological adjustments.

A further consideration concerns the representation of the peat pore structure. The peat is modelled as a single continuum with retention properties prescribed from field

measurements of Finnish peatlands (Menberu et al. 2021). In forestry-drained and degraded peat, structural changes, including cracking, shrinkage, and biologically mediated macropore development, can promote preferential flow that a single-continuum description does not represent explicitly. At the catchment scale, and for the saturated WT dynamics central to this study, these effects are absorbed into the calibrated effective hydraulic conductivity, which reproduces the observed WT behaviour across the monitoring network. A dual-continuum or dual-permeability formulation, available within HydroGeoSphere, would provide a more mechanistic representation of matrix and macropore flow in the degraded peat profile. Such a formulation would, however, require additional macropore-domain parameters, including pore geometry and matrix-macropore exchange, that cannot be constrained without dedicated field characterisation such as tension-infiltration and tracer experiments. Targeted measurement of preferential flow, coupled with a dual-permeability formulation, is therefore a valuable direction for future refinement of the peat-profile representation.

Third, restoration measures are represented in a spatially aggregated and geometrically idealised manner. This is a necessary and appropriate choice at catchment scale, but the continuous zone-based parameterisation assigns uniform hydraulic properties across all features of the same type, and therefore cannot capture the natural variability between individual structures. In practice, peat dams differ in compaction and consolidation state, and log-filled ditches vary in wood alignment, decay stage, and sediment infill; each combination producing its own pattern of preferential flow paths. This structure-to-structure variability is not represented in the current framework, and there are no meaningful ways to constrain the parameters related to restoration activities by field measurements. It is expected that hydraulic responses around individual features are likely more spatially heterogeneous in reality than the model suggests.

Fourth, simulated groundwater recovery reflects a superimposed, landscape-integrated response of multiple interacting restoration measures. This aggregation obscures the independent hydraulic signature of individual features and may conflate their spatial influence radii. As a result, the distance-decay relationships reported in Fig. 11 should be interpreted as emergent landscape-scale patterns rather than the isolated response of any single feature type. Disentangling these contributions would require counterfactual simulations with selective feature deactivation, which could be a productive direction for future design-optimisation studies.

Finally, the simulation encompasses a single hydrological year (2022–2023), representing one realisation of hydroclimatic forcing within the observed variability range at Matorova. Interannual variability in snowpack, autumn precipitation timing, and summer drought intensity may substantially modulate the magnitude and spatial footprint

of restoration-induced groundwater recovery in ways not captured here. Ensemble simulations across multiple forcing years would provide a more robust characterisation of restoration performance under variable climate conditions, and represent a natural next step for further studies as the post-restoration monitoring record grows.

Notwithstanding these constraints, the identified process controls and characteristic response geometries are mechanistically grounded and arise from fundamental hydrological principles rather than site-specific conditions, supporting their transferability across boreal fen systems with comparable peat architecture. Their quantitative expression, however, remains conditioned on the parameterisation, restoration design, and climatic conditions of Matorovasuo, and should be treated as order-of-magnitude benchmarks rather than universal constants. Field-based validation, through multi-year piezometric monitoring and targeted tracer or geophysical surveys, remains essential to test and refine the process hypotheses generated by this framework.

Conclusions

This study identifies characteristic spatial response lengths, governing controls, and process-level mechanisms of restoration-induced WT recovery at the catchment scale. Using calibrated, fully integrated, 3D physics-based modelling, it is shown that rewetting interventions, including ditch infilling and damming, do more than merely elevate the mean WT; they fundamentally reorganise the spatial and temporal structure of peatland hydrology.

Restoration effects propagate well beyond individual interventions, producing substantial gains in nominally undisturbed areas (i.e. areas previously thought unaffected), emphasising the importance of accounting for drainage legacy and lateral hydrological connectivity in restoration planning. Seasonal modulation of these effects further suggests that restoration efficiency measured in WT recovery, is not constant through time. Wet periods amplify gains, producing increases nearly twice those observed during late-winter low-flow conditions, while elevated efficiency persists through dry spells between spring melts and autumn rains. This highlights the importance of considering seasonal hydrological regimes and year-to-year variability when evaluating restoration performance and setting monitoring expectations.

Importantly, the extent and spatial footprint of groundwater recovery depend strongly on restoration structure type and local peatland context. Low-permeability peat infillings produce pronounced but spatially limited responses, whereas dams generate broader, more diffuse impacts, albeit with greater variability. From a

management perspective, these findings imply trade-offs between localised control and wider-area influence that should be explicitly considered when selecting restoration strategies and their spatial configuration. Restoration effectiveness is strongly conditioned by internal peatland geomorphology, with maximum groundwater recovery occurring in interior areas of intermediate peat thickness and attenuation near peat–mineral boundaries where lateral leakage dominates.

The principal contribution of this work is a transferable, mechanistic synthesis grounded in fully integrated hydrogeological modelling that links process-based controls to design-relevant geometry. While the quantitative magnitudes reported here are conditioned on the hydrogeological setting, restoration design, and climate of Matorovasuo, the governing mechanisms, connectivity-driven redistribution, scale-dependent attenuation, and geomorphological control, arise from general hydrological principles and are therefore expected to transfer across comparable boreal fen systems. On this basis, the results offer a framework to anticipate where, when, and why peatland restoration interventions will be hydrologically effective, drawing on characteristic response lengths, connectivity dynamics, and geomorphic constraints. By resolving how restoration measures interact with peatland structure and hydroclimatic forcing to reshape groundwater connectivity, this study provides a mechanistic basis for more deliberate, efficient, and defensible peatland restoration design.

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Data availability The datasets generated and analysed during the current study are not publicly available but are available from the corresponding author on reasonable request.

Declarations

Ethics approval Not applicable. This study did not involve human or animal subjects, their data, or biological material.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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